Data Mining: Data Preprocessing

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2026/09/18, Week 2

Why data preprocessing?

- Data in the real world is dirty.
- Why is data dirty:
  ① Incomplete
  ② Noisy
  ③ Inconsistent
  ④ Intentional

Why data preprocessing (cont.)

- Incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data.
  - N/A data value when collected.
  - Different consideration between the time when the data was collected and when it is analyzed.
  - Human / Hardware/ Software problems.
  - E.g.,
    ① name = “”
    ② birthday = 2013/02

Why data preprocessing (cont.)

- Noisy (Inaccurate): containing errors or outliers
  - Collection / Entry / Transmission of data
  - E.g.,
    ① age = -10
    ② Salary = -100M

Why data preprocessing (cont.)

- Inconsistent: containing discrepancies in codes or names
  - Different data sources
  - Functional dependency violation
  - E.g.,
    ① birthday = 104/01/05 and 2015/1/5

Why data preprocessing (cont.)

- Intentional:
  - Disguised missing data
  - E.g.,
    ① Feb. 29 as everyone’s birthday

Objective

- To conduct successful data mining, the first important thing is to get familiar with your data
  - ① What are the types of attributes or fields that make up your data?
  - ② What kind of values does each attribute have?
  - ③ How are the values distributed?
  - ④ How can we measure the similarity of some data objects with respect to others? (discuss this in clustering)
- No quality data, no quality mining results
  - ① How can we measure the quality of data?
  - ② How can we clean and integrate data from multiple heterogeneous sources?
  - ③ How can we normalize, compress, or transform the data?
  - ④ How can we reduce the dimensionality of data to help subsequent analysis?

Data set types

① Record Data
② Graphs and Networks
③ Ordered Data
④ Spatial, image and multimedia Data

① Record Data

- Relational tables, highly structured

Person:

Pers_ID	Surname	First_Name	City
0	Miller	Paul	London
1	Ortega	Alvaro	Valencia
2	Huber	Urs	Zurich
3	Blanc	Gaston	Paris
4	Bertolini	Fabrizio	Rom

no relation

Car:

Car_ID	Model	Year	Value	Pers_ID
101	Bentley	1973	100000	0
102	Rolls Royce	1965	330000	0
103	Peugeot	1993	500	3
104	Ferrari	2005	150000	4
105	Renault	1998	2000	3
106	Renault	2001	7000	3
107	Smart	1999	2000	2

① Record Data (cont.)

- Transaction data

TID	*Items*
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

① Record Data (cont.)

- Data matrix, e.g., numerical matrix, crosstabs

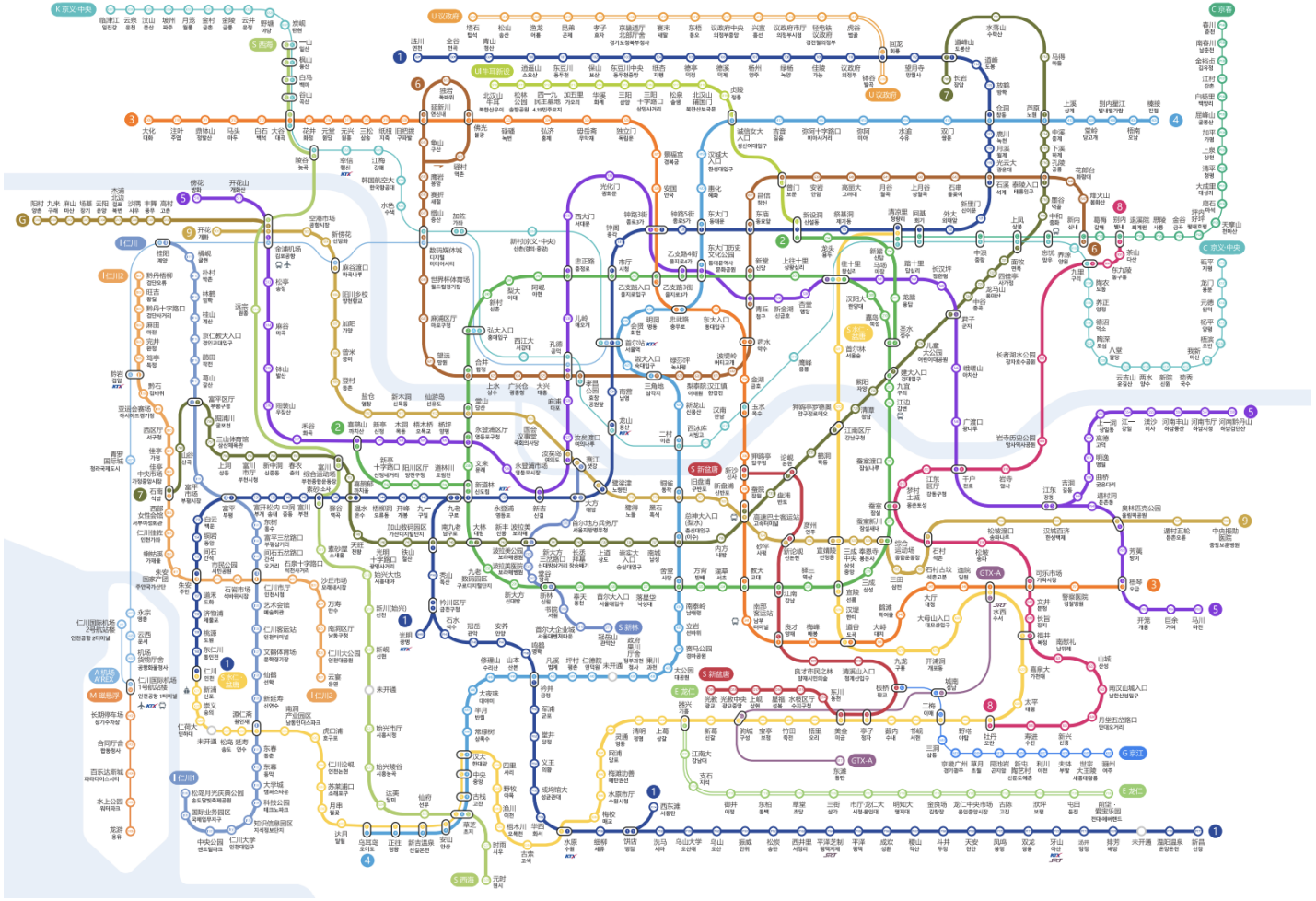
	China	England	France	Japan	USA	Total
Active Outdoors Crochet Glove		12.00	4.00	1.00	240.00	257.00
Active Outdoors Lycra Glove		10.00	6.00		323.00	339.00
InFlux Crochet Glove	3.00	6.00	8.00		132.00	149.00
InFlux Lycra Glove		2.00			143.00	145.00
Triumph Pro Helmet	3.00	1.00	7.00		333.00	344.00
Triumph Vertigo Helmet		3.00	22.00		474.00	499.00
Xtreme Adult Helmet	8.00	8.00	7.00	2.00	251.00	276.00
Xtreme Youth Helmet		1.00			76.00	77.00
Total	14.00	43.00	54.00	3.00	1,972.00	2,086.00

Types of Data Sets: ① Record Data

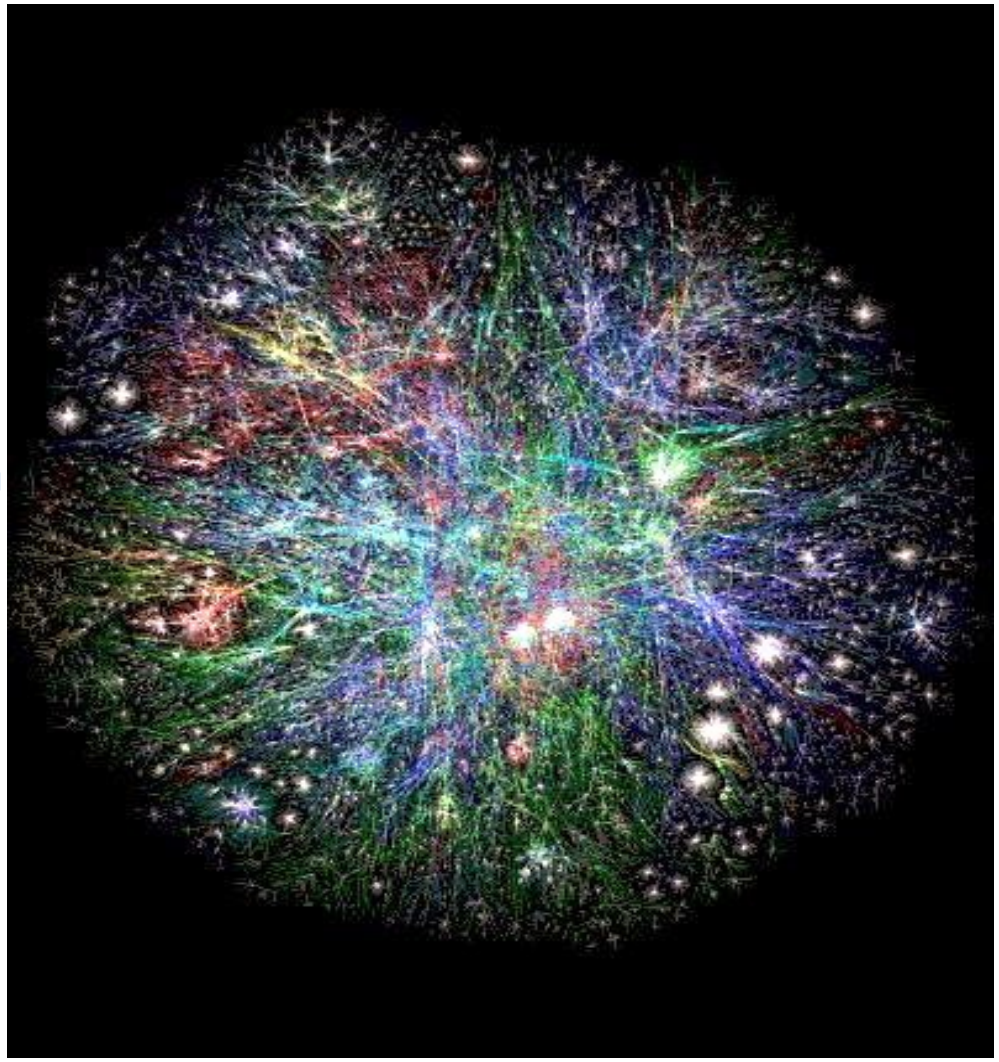
- Document data: Term-frequency vector (matrix) of text documents

	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

② Graphs and Networks

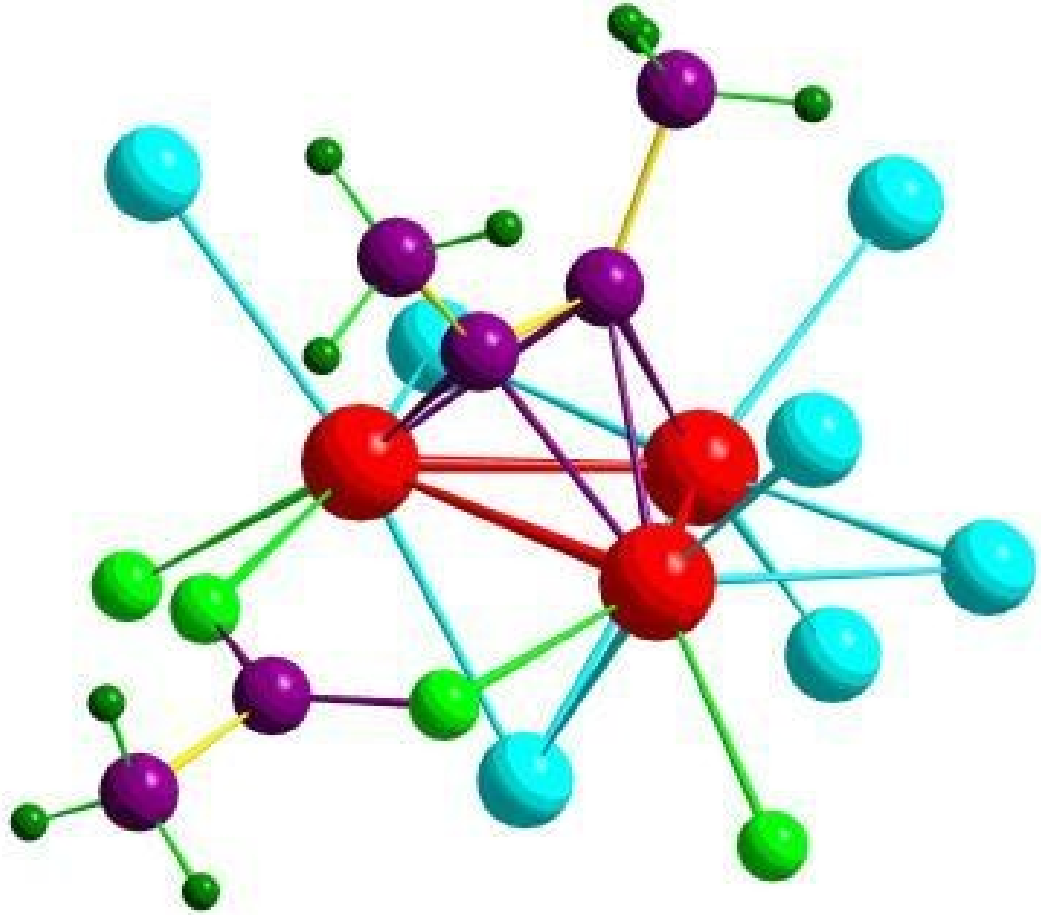


Transportation network

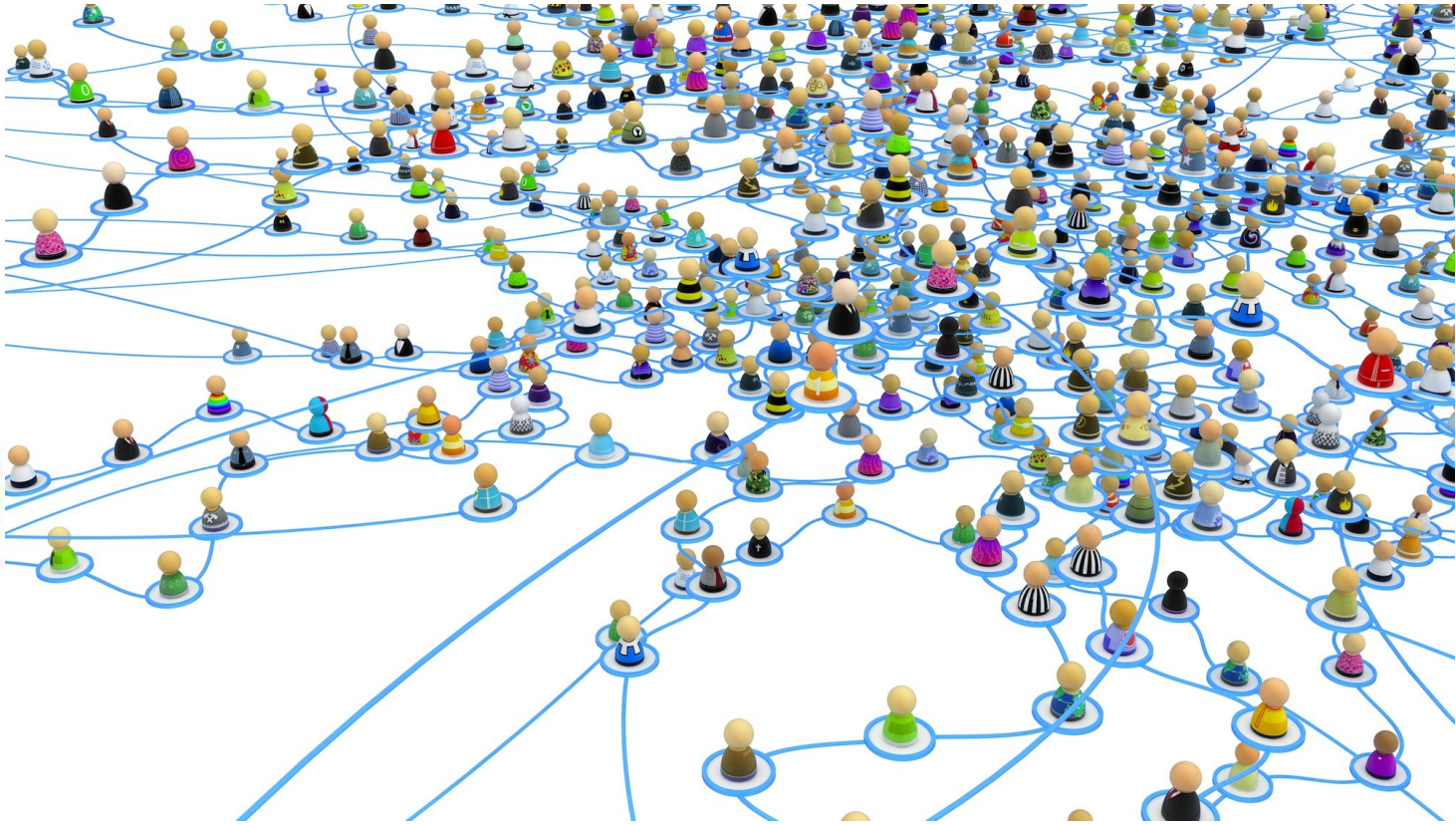


World Wide Web

② Graphs and Networks (cont.)



Molecular Structures



Social or information networks

③ Ordered Data

- Video data: sequence of images



③ Ordered Data (cont.)

- Temporal data: time-series



開盤	2,415.00	市值	62.50兆	股息	1.00%
最高	2,415.00	本益比	27.58	季度股利金額	6.02
最低	2,390.00	52 週高點	2,535.00	52 週低點	1,160.00

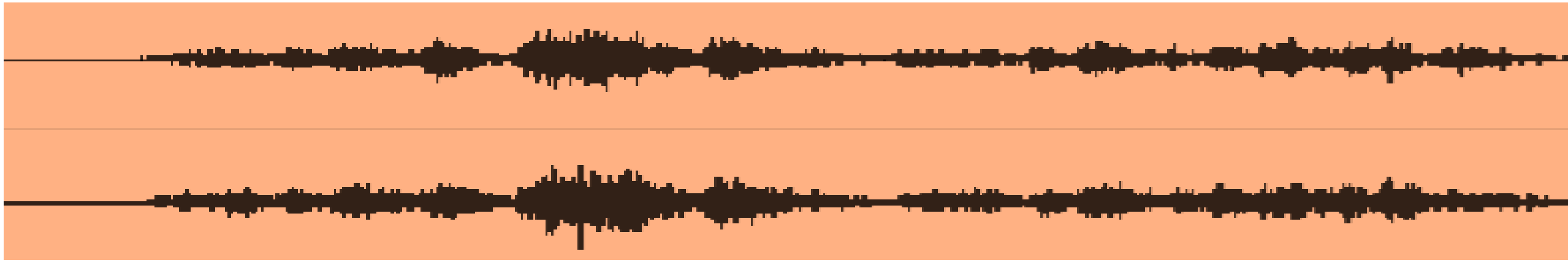
③ Ordered Data (cont.)

- Genetic sequence data



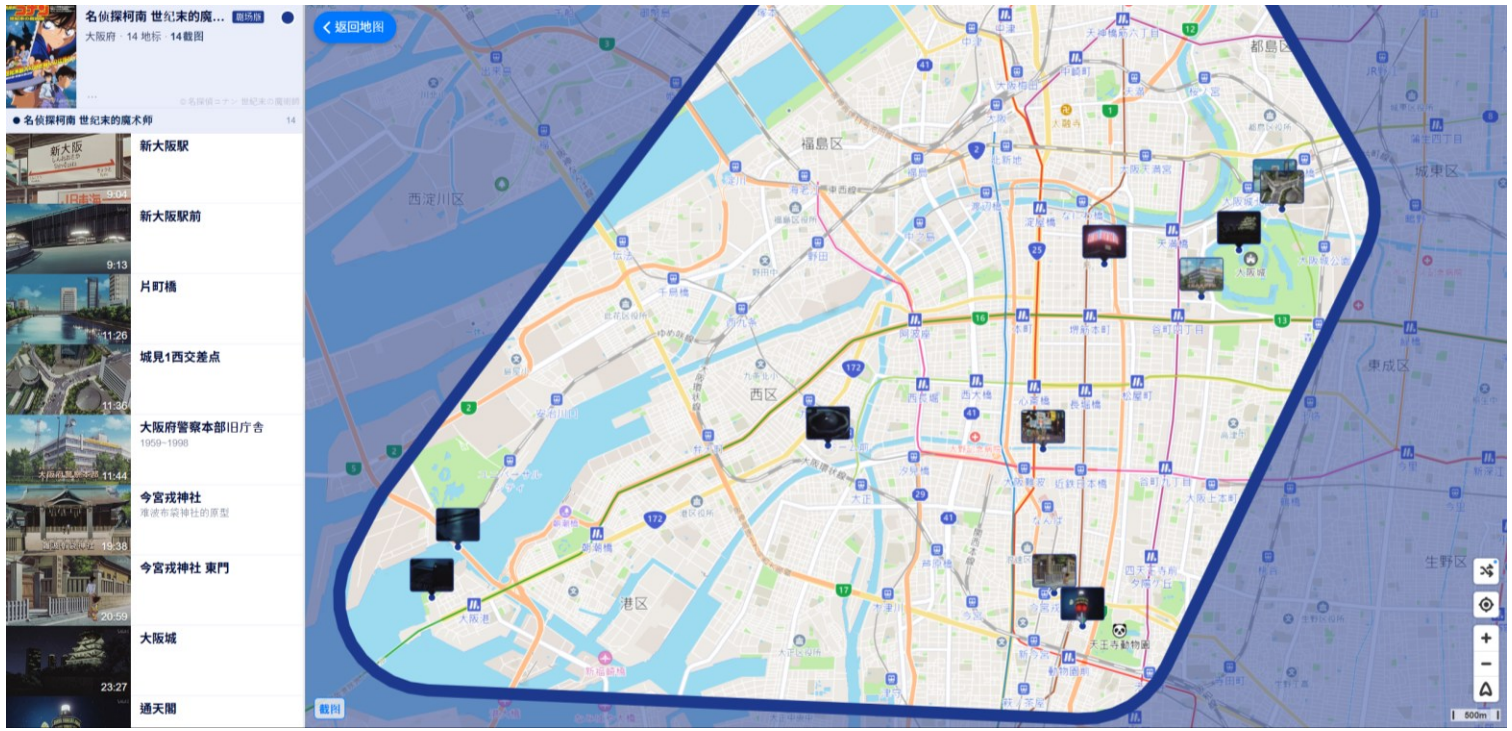
③ Ordered Data

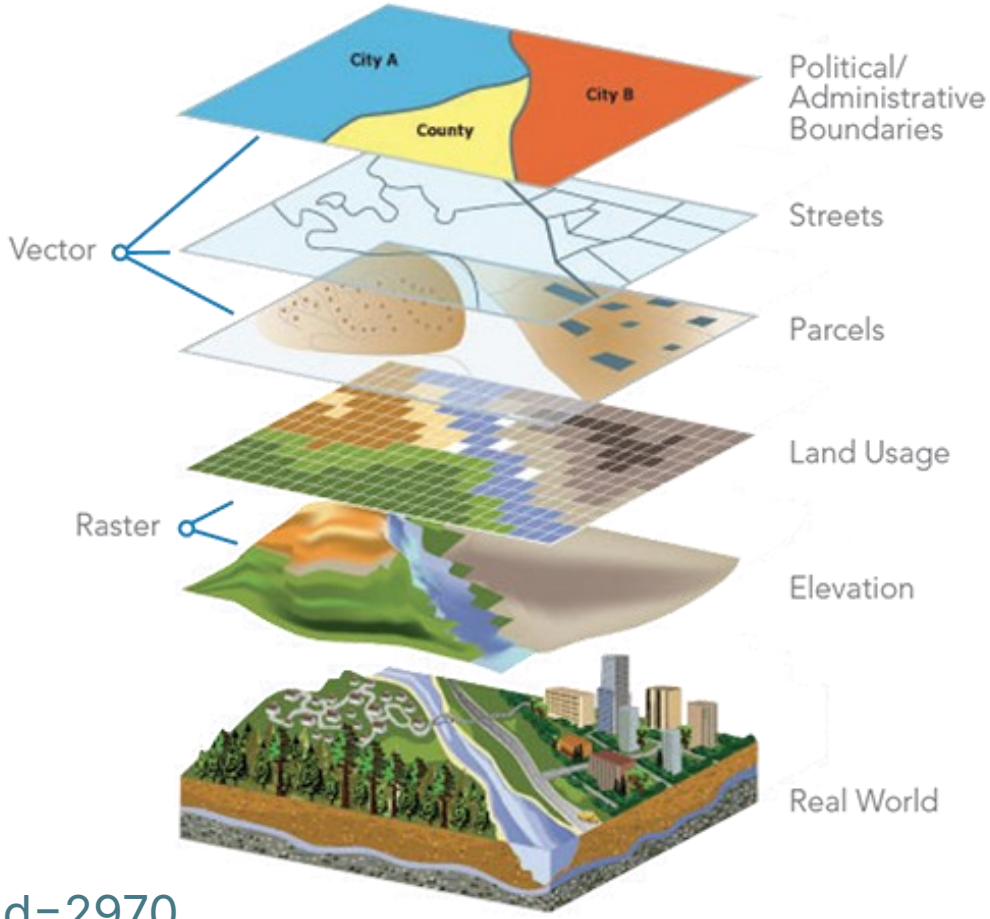
- Audio Signal



Spatial, image and multimedia Data

- Spatial data: maps





Source:
https://www.anitabi.cn/map?c=135.4853,34.6771&z=12.7&bangumiId=2970

Data types

The various attribute types

Data objects and Attributes

- A dataset is a collection of data objects.
- A data object represents an entity or observation.
  - samples, examples, instances, or datapoints
- An attribute describes a characteristic of a data object.
  - dimension, feature, or variable



Data Object

Attribute

Source:呪術廻戦公式ファンブック

The type of an attribute

- The type of an attribute is determined by the set of possible values.
  ① Nominal: red, blue (Colors)
  ② Binary: true, false (Boolean)
  ③ Ordinal: freshman, sophomore, junior, senior (grade)
  ④ Numeric (Quantitative)
    - Interval-scaled vs. Ratio-scaled
    - Discrete vs. Continuous Attributes

① Nominal attributes

- Nominal means relating to names
  - Symbols or names of things
  - Some kind of category, code, or state
  - Nominal attributes are also referred to as Categorical
- hair_color and marital_status are two attributes describing person objects
  - hair_color: can be black, brown, blond, red, auburn, gray, and white
  - marital_status: can be on single, married, divorced, and widowed

① Nominal attributes (cont.)

- Nominal attribute values do not have any meaningful order about them and are not quantitative
  - No sense to find the mean (average) value or median (middle) value

② Binary attributes

- Binary attribute is a nominal attribute with only two categories or states: 0 (absent) or 1 (present)
  - Boolean➔ 0: False (Negative), 1: True (Positive)
- Symmetric vs Asymmetric

② Binary attributes (cont.)

1) Symmetric: both of its states are equally valuable and carry the same weight
   - The attribute smoker describing a patient object
     - 1 indicates that the patient smokes
     - 0 indicates that the patient does not

② Binary attributes (cont.)

2) Asymmetric: the outcomes of the states are not equally important, such as the positive and negative outcomes
   - The patient undergoes a medical test that has two possible outcomes
     - 1: means the result of the test for the patient is positive.
     - 0: means the result is negative.

③ Ordinal attributes

- Ordinal attribute is an attribute with possible values that have a meaningful order or ranking
  - The differences between categories are not necessarily equal or quantitatively meaningful.
  - The size of drinks available: small, medium, and large
  - Grade: A+, A, A−, B+, B, B-, …
- Ordinal data tell us which is higher, but not how much higher.

③ Ordinal attributes (cont.)

- Ordinal attributes are useful for recording ordered categories, judgments, or ratings.
  - Surveys for ratings, 5-point Likert Scale
    - (1) very dissatisfied, (2) dissatisfied, (3) neutral (4) satisfied, and (5) very satisfied
- Central tendency of an ordinal attribute can be represented by its mode and its median , but the mean cannot be defined

貓界鄙視鏈



④ Numeric attributes

- A numerical attribute represents a measurable or countable quantity using numerical values.
- Numerical attributes can be classified by:
  - Interval-scaled vs Ratio-scaled
    - Temperature: Celsius and Fahrenheit (Interval), Kelvin (Ratio)
  - Discrete vs Continuous

Interval-scaled vs. Ratio-scaled

- Interval-scaled attributes
  - Equal numerical differences represent equal changes.
  - The zero point is arbitrary and does not indicate the absence of the quantity.
- Ratio-scaled attributes
  - A true zero represents the absence of the measured quantity.
  - If a measurement is ratio-scaled, we can speak of a value as being a multiple (or ratio) of another value.

Discrete vs. Continuous Attributes

- Discrete Attribute
  - Take a finite or countably infinite set of distinct values.
  - Values are usually obtained by counting or labeling.
    - Numerical examples: number of students, number of purchases
    - Categorical examples: ZIP code, profession, words in documents
  - May be represented by integers, but integer representation does not necessarily imply numerical meaning.
  - Binary attributes are a special case of discrete attributes.

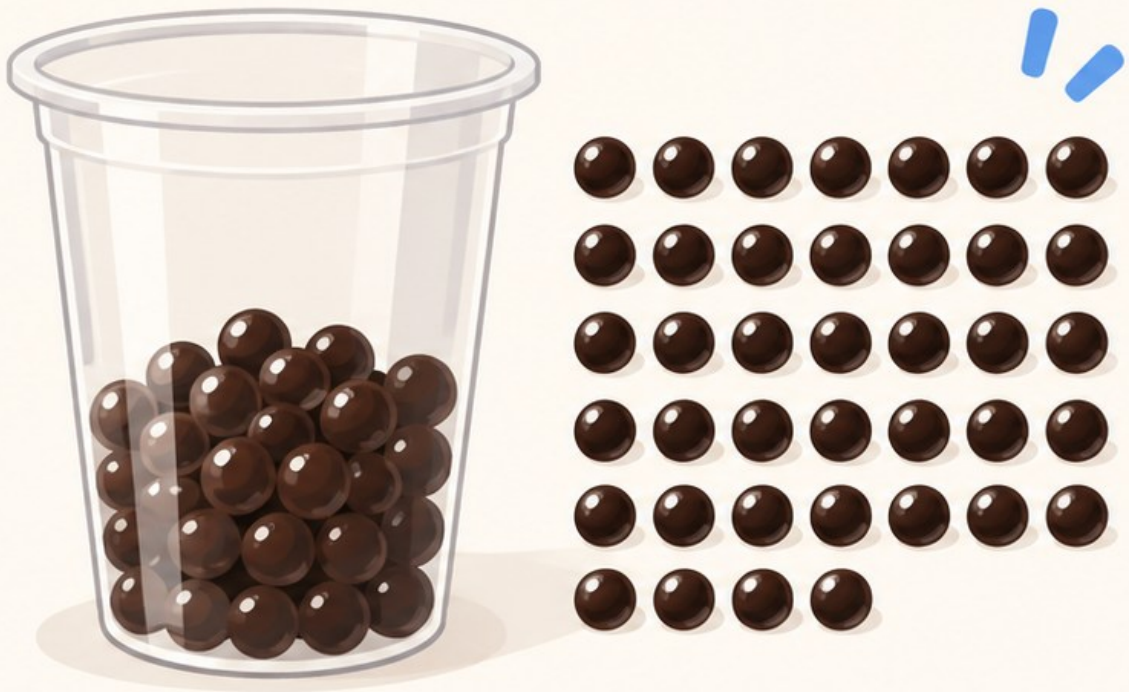
Takeaway: Discrete values are countable and separated.

Discrete vs. Continuous Attributes

- Continuous Attribute
  - Can theoretically take any real value within an interval.
  - Values are usually obtained by measurement.
    - E.g., temperature, height, weight, and elapsed time
  - Measurement precision is finite in practice.
  - Usually represented using floating-point numbers.

Takeaway: Continuous values are measurable and can vary smoothly.

Discrete



38 pearls

Counted

Continuous



38.5 g

Measured

Types of attributes:
①Nominal
②Binary
③Ordinal
④Numeric

姓名: 五條悟
年齡: 28歲
誕生日: 12/7
等級: 特級咒術師
高專入學方法: 家系
術式: 無下限咒術



Statistics of data

① Measuring the central tendency

② Dispersion, Correlation

③ Graphical Displays of Data

Basic Statistical Descriptions of Data

- Motivation
  - To better understand the data: central tendency, variation and spread
- For data preprocessing to be successful, it is essential to have an overall picture of your data.
  - Basic statistical descriptions can be used to identify properties of the data.

Statistics of data (cont.)

- Measures of central tendency
  - The location of the middle or center of a data distribution (where do most of the values fall).
- Dispersion of the data
  - How are the data spread out? (distribution)
- Co-variance and correlation coefficient
  - Relations among multiple variables.
- Graphical Displays of Data

① Measuring the central tendency

- Mean (算術平均數)
- Median (中位數)
- Mode (眾數)

Mean (arithmetic mean)

- Common and effective numeric measure of the "center" of a set of data.
  - $x_1 + x_2 + \cdots + x_N$ be a set of *N* values or observations, such as for some numeric attribute X

$$\bar{x} = \frac{\sum_{i=1}^{N} x_i}{N} = \frac{x_1 + x_2 + \cdots + x_N}{N} = \mu$$

Weighted average

- Sometimes, each value x_i in a set may be associated with a weight $w_i, for\ i = 1, \ldots, N$
- Weights reflect the significance, importance, or occurrence frequency attached to their respective value.

$$\bar{x} = \frac{\sum_{i=1}^{N} w_i x_i}{\sum_{i=1}^{N} w_i} = \frac{w_1 x_1 + w_2 x_2 + \cdots + w_N x_N}{w_1 + w_2 + \cdots + w_N}$$

Mean (cont.)

- Although the mean is the single most useful quantity for describing a data set, it is not always the best way of measuring the center of the data.
  - The mean is its sensitivity to extreme (e.g., outlier) values
  - The mean salary between me and Terry Gou
- Trimmed mean: the mean obtained after chopping off values at the high and low extremes.

Median

- For skewed (asymmetric) data, a better measure of the center of data is the median, which is the middle value in a set of ordered data values
  - Given data set of *N* values for an attribute X is sorted in ascending order
    - N is odd: the median is the middle value of the ordered set
    - N is even: the median is not unique; it is the two middlemost values and any value in between

Median (cont.)

- For intervals

Sum before the median interval

$$median \approx L_1 + \left(\frac{N/_2 - (\sum freq)_l}{freq_{median}}\right) \times width$$

Low interval limit

Interval width ($L_2 - L_1$)

$$median = 21 + \frac{\frac{3194}{2} - 950}{1500} \times (50 - 21)$$

age	$frequency$
1–5	200
6–15	450
16–20	300
21–50	1500
51–80	700
81–110	44

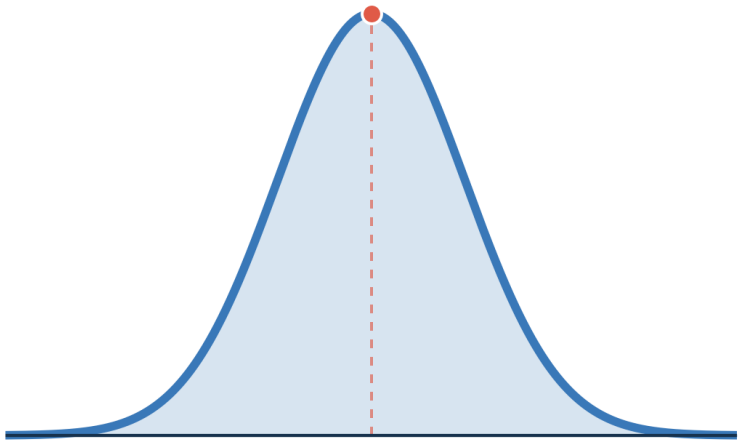
Total: 3194

Mode

- The mode is the value that occurs most frequently in a dataset.
  - Determined for qualitative and quantitative attributes.
  - Unimodal: one mode
    - Empirical formula: $mean - mode = 3 \times (mean - median)$
  - Bimodal: two modes
  - Multimodal: more than two modes
  - No mode: no value occurs more frequently than the others.

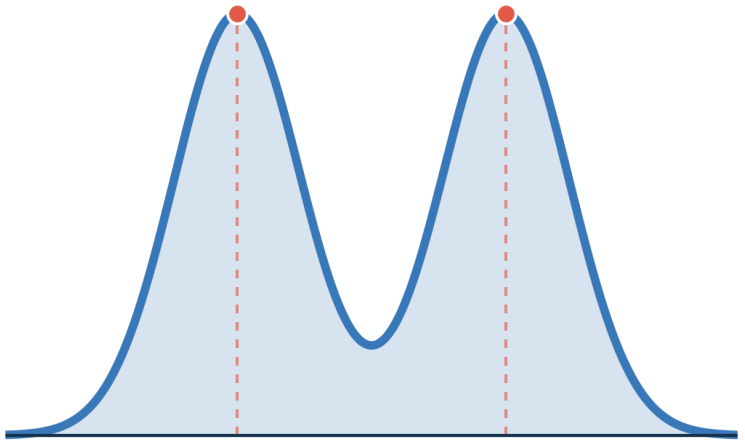
Mode (cont.)

Unimodal



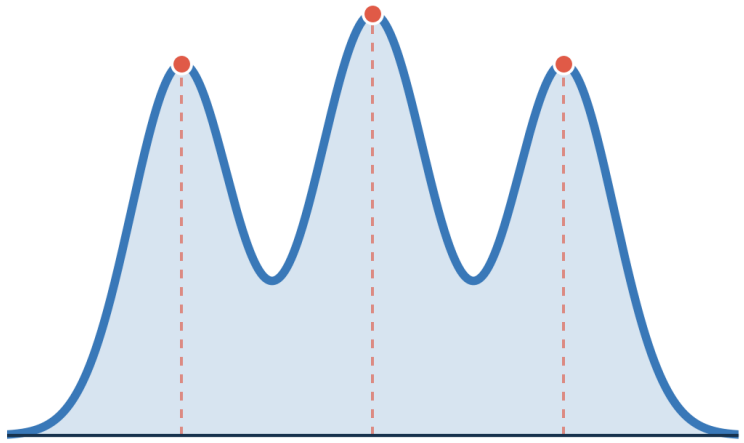
One mode

Bimodal



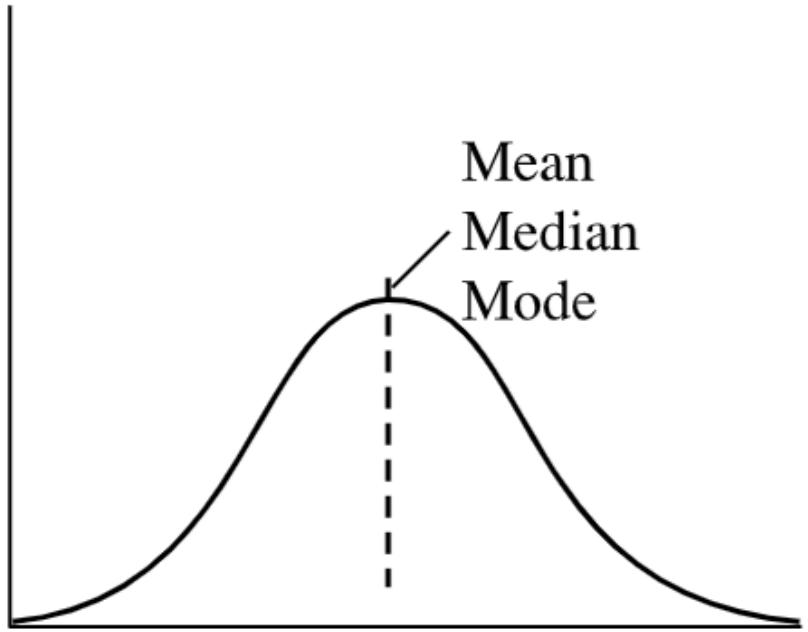
Two modes

Multimodal

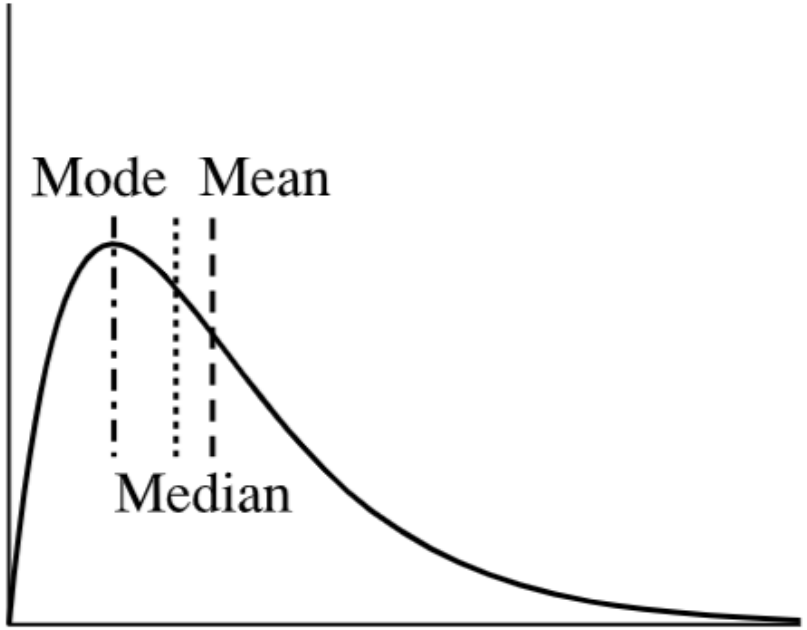


Multiple modes

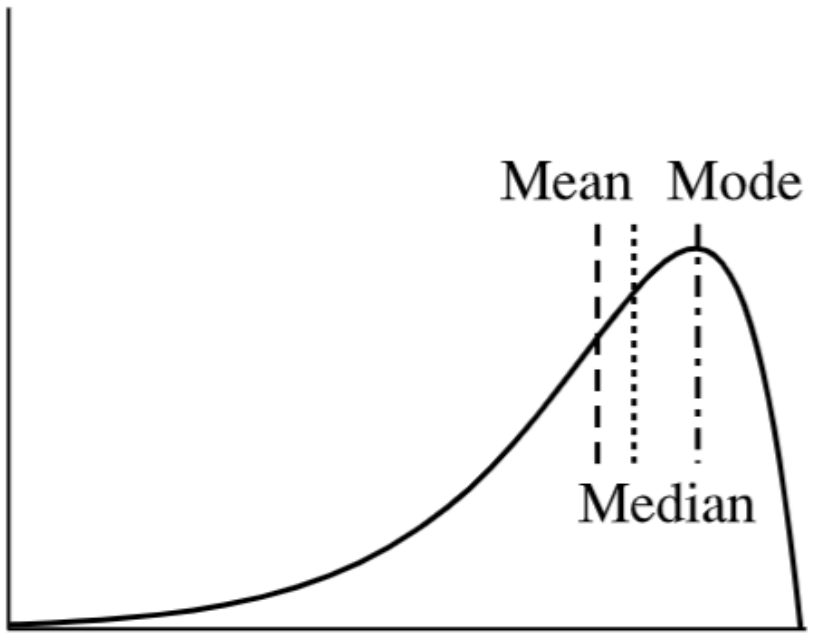
Symmetric vs. Skewed Data



(a) Symmetric data



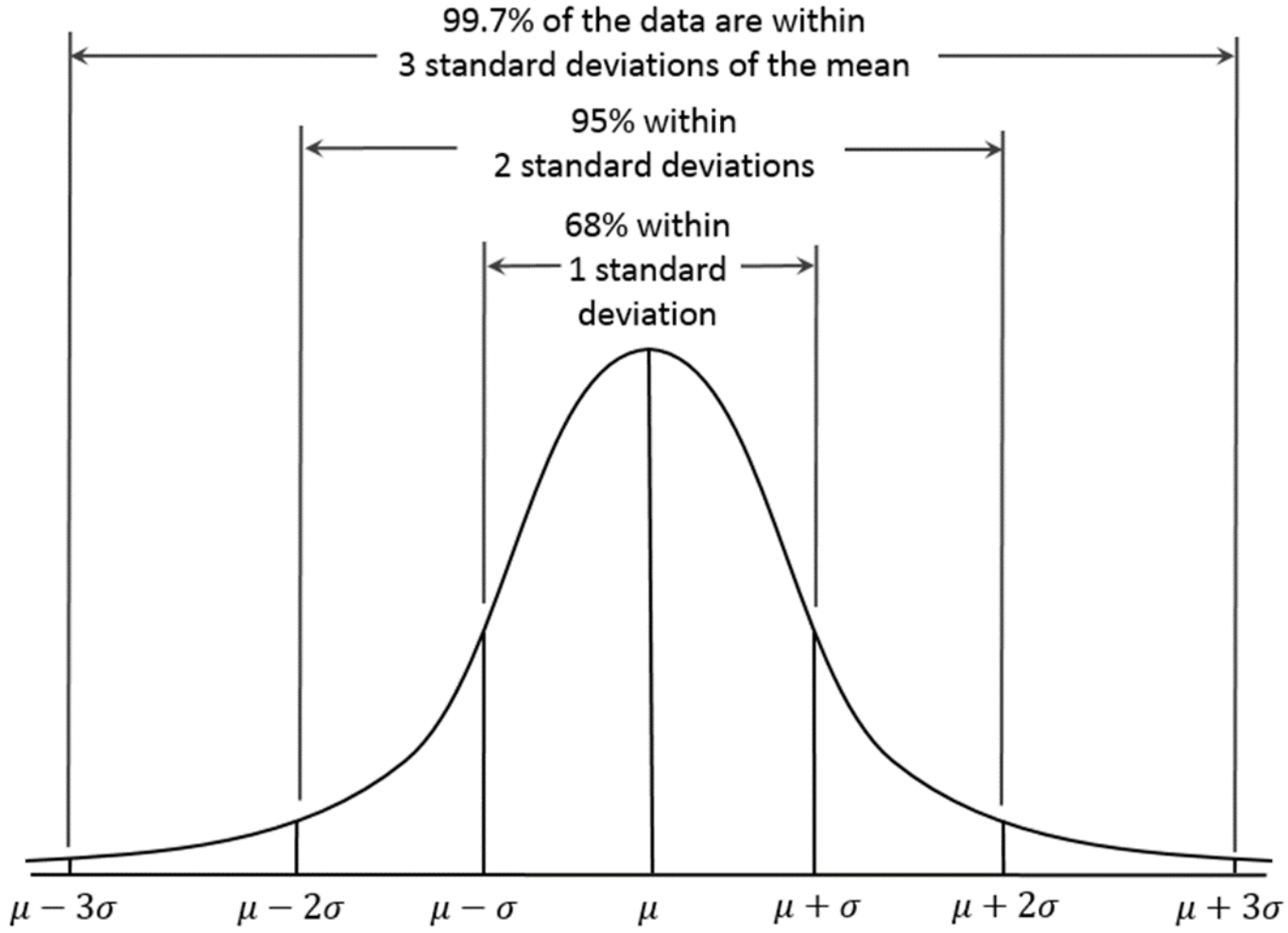
(b) Positively skewed data



(c) Negatively skewed data

Properties of Normal Distribution Curve

← — ————Represent data dispersion, spread — ————→



Represent central tendency

② Measuring The Dispersion of Data

- Variance and standard deviation (變異數與標準差)
  - Covariance
  - Correlation
- Range (全距)
- Quantiles (分位數)
- Quartiles (四分位數)
- Percentiles (百分位數)
- Interquartile range (四分位距)

Variance and Standard Deviation

- Variance and standard deviation (sample: s, population: σ)
  - Measure how far data values typically spread from the mean.
    - Low standard deviation: values are concentrated near the mean.
    - High standard deviation: values are spread farther from the mean.
  - Sample variance: $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$
  - Population variance: $\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2$
- Standard deviation s (or σ) is the square root of variance s^2 (or σ^2)

n : the size of the sample
N : the size of the population

Variance for Single Variable (Numerical Data)

- The variance of a random variable X provides a measure of how much the value of X deviates from the mean or expected value of X.

$$\sigma^2 = \mathrm{var}(X) = E[(\mathrm{X}-\mu)^2] = \begin{cases} \sum_x (x-\mu)^2 f(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} (x-\mu)^2 f(x)dx & \text{if } X \text{ is continuous} \end{cases}$$

- Expect value of X: $\mu = \mathrm{E}[\mathrm{X}]$

$$\sigma^2 = \mathrm{var}(X) = E[(\mathrm{X}-\mu)^2] = E[X^2] - \mu^2 = E[X^2] - (E[X])^2$$

variance = mean of the squares – square of the mean

Covariance of Two Attributes

- How much two attributes change together?
- Covariance measures how two numerical attributes vary together.
- Given n paired observations: $\{(a_1, b_1), \ldots, (a_n, b_n)\}$

$$\bar{a} = \frac{\sum_{i=1}^{n} a_i}{n}, \bar{b} = \frac{\sum_{i=1}^{n} b_i}{n}$$

$$Cov(A, B) = \frac{\sum_{i=1}^{n} (a_i - \bar{a})(b_i - \bar{b})}{n}$$

Covariance of Two Variables

- When X_1, X_2 are treated as random variables:
- Mean or expected value of X_1, X_2

$$\mu_1 = E(\mathrm{X}_1), \mu_2 = E(\mathrm{X}_2)$$

- Covariance between X_1 and X_2

$$Cov(X_1, X_2) = E\big((X_1 - \mu_1)(X_2 - \mu_2)\big) = E(X_1 X_2) - \mu_1 \mu_2$$
$$= E(X_1 X_2) - E[X_1]E[X_2]$$

Interpreting Covariance

- For each observation, examine the product:

$$(a_i - \bar{a})(b_i - \bar{b})$$

  - $Cov(A, B) > 0$: tend to move in the same direction.
  - $Cov(A, B) < 0$: tend to move in the opposite direction.
  - $Cov(A, B) \approx 0$: no clear linear co-movement.
- If A and B are independent $\Rightarrow Cov(A, B) = 0$
  - $Cov(A, B) = 0 \not\Rightarrow A$ and B are independent
  - Zero covariance does not rule out a nonlinear relationship

Covariance Example

$$Cov(A,B) = \frac{\sum_{i=1}^{n}(a_i - \bar{A})(b_i - \bar{B})}{n}$$

- Suppose two stocks X_1 and X_2 have the following values in one week:
  - (2, 5), (3, 8), (5, 10), (4, 11), (6, 14)
- Question: If the stocks are affected by the same industry trends, will their prices rise or fall together?
  - $\overline{X_1} = \frac{2+3+5+4+6}{5} = 4,\ \overline{X_2} = \frac{5+8+10+11+14}{5} = 9.6$
  - $Cov(X_1, X_2) = \frac{(2-4)(5-9.6)+(3-4)(8-9.6)+(5-4)(10-9.6)+(4-4)(11-9.6)+(6-4)(14-9.6)}{5} = 4$
- X_1 and X_2 rise together (positive covariance) since $Cov(X_1, X_2) > 0$

Correlation Coefficient for Numeric

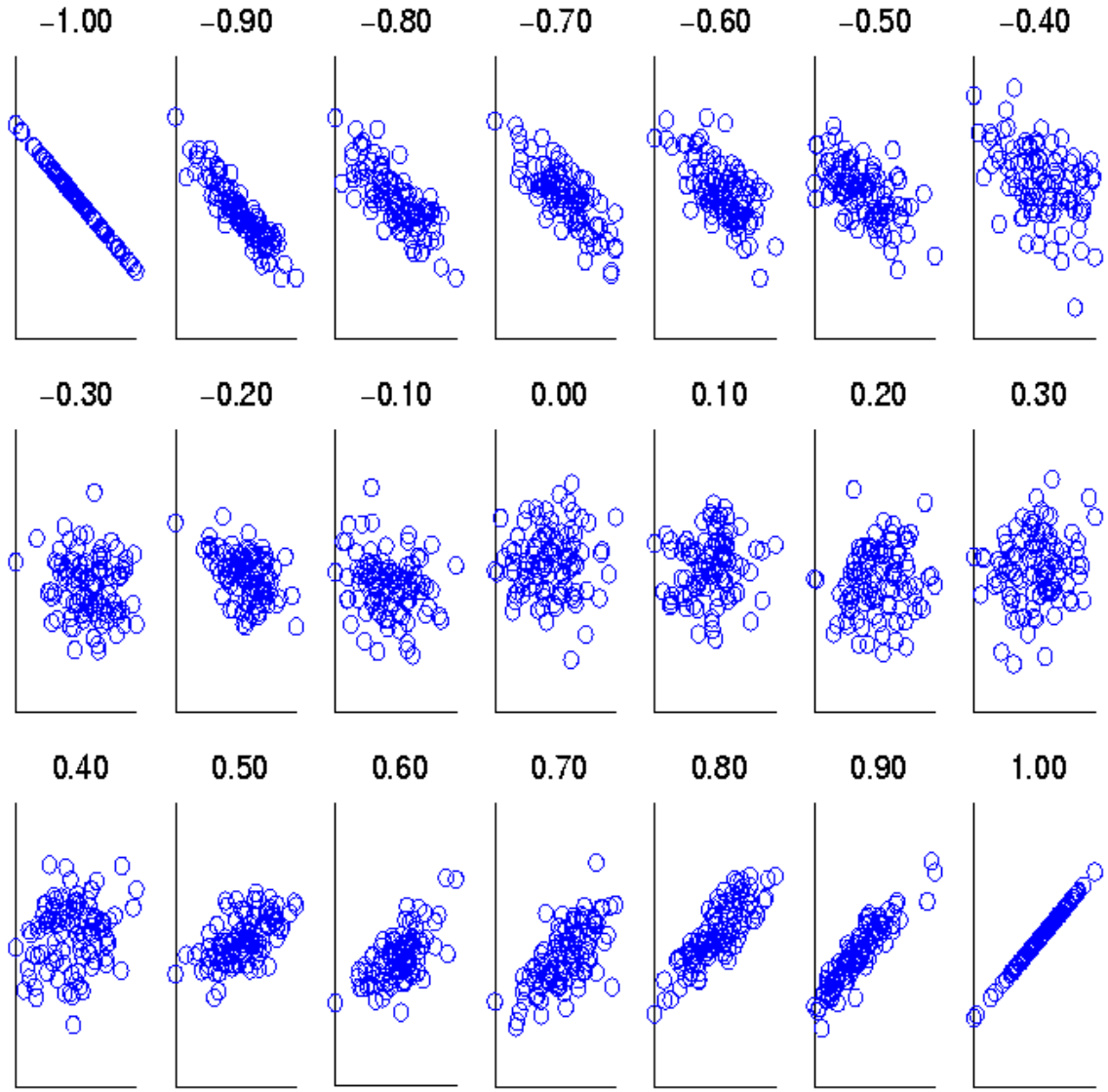
- Correlation between two variables X_1 and X_2 is the standard covariance, obtained by normalizing the covariance with the standard deviation of each variable

$$r_{A,B} = \frac{Cov(A,B)}{Std(A)Std(B)}$$

  - Covariance: the tendency in the linear relationship between the variables
  - Correlation: Dividing the standard deviations to normalize the measure, creating a unit-less
  - $1 \geq r > 0$: positively correlated
  - $r = 0$: independent
  - $-1 \leq r < 0$: negative correlated

If A and B are correlated, this does not necessarily imply that A causes B or that B causes A

Changes of Correlation Coefficient



Covariance Matrix

$$\Sigma = \begin{array}{c} \\ X_1 \\ X_2 \\ X_3 \end{array} \begin{array}{c} \begin{array}{ccc} X_1 & X_2 & X_3 \end{array} \\ \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_2^2 & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_3^2 \end{bmatrix} \end{array}$$

Diagonal: variance of each variable

$$\sigma_i^2 = \mathrm{Var}(X_i)$$

Off-diagonal: covariance between variables

$$\sigma_{ij} = \mathrm{Cov}(X_i, X_j)$$

Symmetric matrix

$$\mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i)$$

Correlation Test for Nominal (Categorical)

- Tests two categorical attributes are associated can be discovered by a X^2(chi-square) test
  - ➢ H_0: A and B are independent
  - ➢ H_1: A and B are associated

	Play chess	Not play chess	Sum (row)
Like science fiction	250 (X1)	200 (X2)	450
Not like science fiction	50 (X3)	1000 (X4)	1050
Sum(col.)	300	1200	1500

Contingency Table

- Expected frequency :

$$e_{ij} = \frac{count(i) \times count(j)}{total\ number} \Rightarrow e_{11} = \frac{300 \times 450}{1500}$$

- X^2(chi-square) test

$$X^2 = \sum_i \sum_j \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

Chi-Square Example

- $\chi^2 = \frac{(250-90)^2}{90} + \frac{(50-210)^2}{210} + \frac{(200-360)^2}{360} + \frac{(1000-840)^2}{840} = 507.93$

a) $e_{11} = \frac{450 \times 300}{1500} = 90$

b) $e_{21} = \frac{1050 \times 300}{1500} = 210$

c) $e_{12} = \frac{450 \times 1200}{1500} = 360$

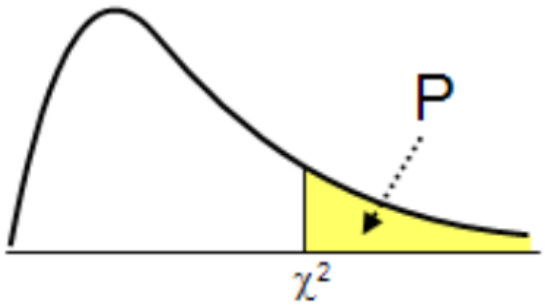
d) $e_{22} = \frac{1050 \times 1200}{1500} = 840$

	Play chess	Not play chess	Sum (row)
Like science fiction	250 (X1)	200 (X2)	450
Not like science fiction	50 (X3)	1000 (X4)	1050
Sum(col.)	300	1200	1500

Chi-Square Example

Values of the Chi-squared distribution

It shows that like_science_fiction and play_chess are correlated in the group



$$\chi^2 = \frac{(250-90)^2}{90} + \frac{(50-210)^2}{210} + \frac{(200-360)^2}{360} + \frac{(1000-840)^2}{840} = 507.93$$

We can reject the null hypothesis of independence at a confidence level of 0.001

Reject H_0

	P										
DF	**0.995**	**0.975**	**0.20**	**0.10**	**0.05**	**0.025**	**0.02**	**0.01**	**0.005**	**0.002**	**0.001**
1	0.0000393	0.000982	1.642	2.706	3.841	5.024	5.412	6.635	7.879	9.550	10.828
2	0.0100	0.0506	3.219	4.605	5.991	7.378	7.824	9.210	10.597	12.429	13.816
3	0.0717	0.216	4.642	6.251	7.815	9.348	9.837	11.345	12.838	14.796	16.266
4	0.207	0.484	5.989	7.779	9.488	11.143	11.668	13.277	14.860	16.924	18.467
5	0.412	0.831	7.289	9.236	11.070	12.833	13.388	15.086	16.750	18.907	20.515
6	0.676	1.237	8.558	10.645	12.592	14.449	15.033	16.812	18.548	20.791	22.458

Degree of freedom (#categories_in_variable_A - 1)(#categories_in_variable_B - 1)

- number of values that are free to vary

Measuring The Dispersion of Data

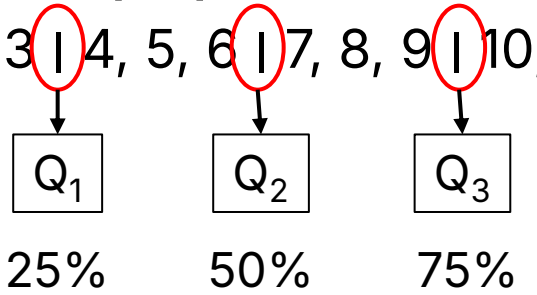
- Range
  - The difference between the largest and smallest values

$$Range = x_{max} - x_{min}$$

  - Simple to calculate, but sensitive to outliers.
- Quantiles
  - Quantiles are cut points that divide ordered data into approximately equal-sized groups.
  - The kth q-quantile corresponds to the cumulative proportion:

$$\frac{k}{q}, k = 1,2, \dots, q-1$$

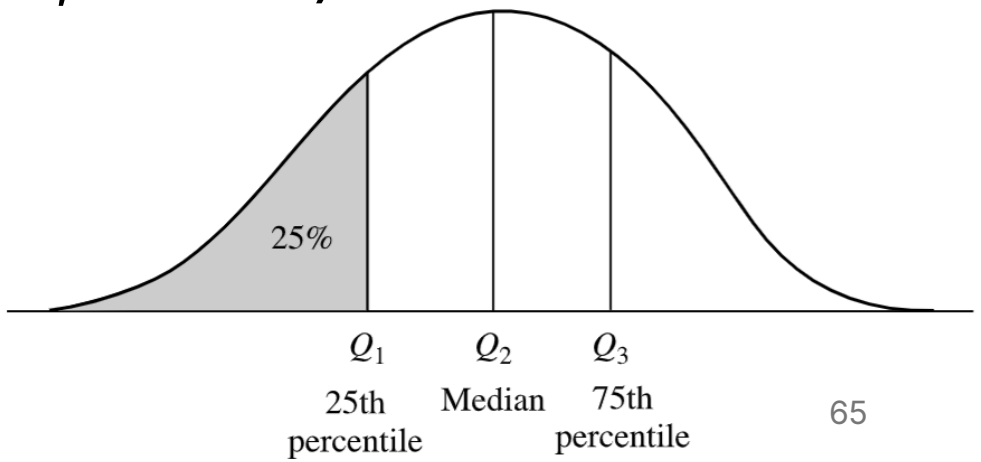
    - 1, 2, 3 | 4, 5, 6 | 7, 8, 9 | 10, 11, 12



Quantiles divide the ordered data into equal-sized groups.

Measuring The Dispersion of Data

- Quartiles (4-quantiles)
  - Three cut points that divide ordered data into four approximately equal-sized groups.
  - Each part represents one-fourth of the data distribution
  - For q=4, there are q−1=3 quartile cut points.
    - First quartile: (Q1, 25th percentile)
    - Second quartile: (Q2, 50th percentile, median)
    - Third quartile: (Q3, 75th percentile)



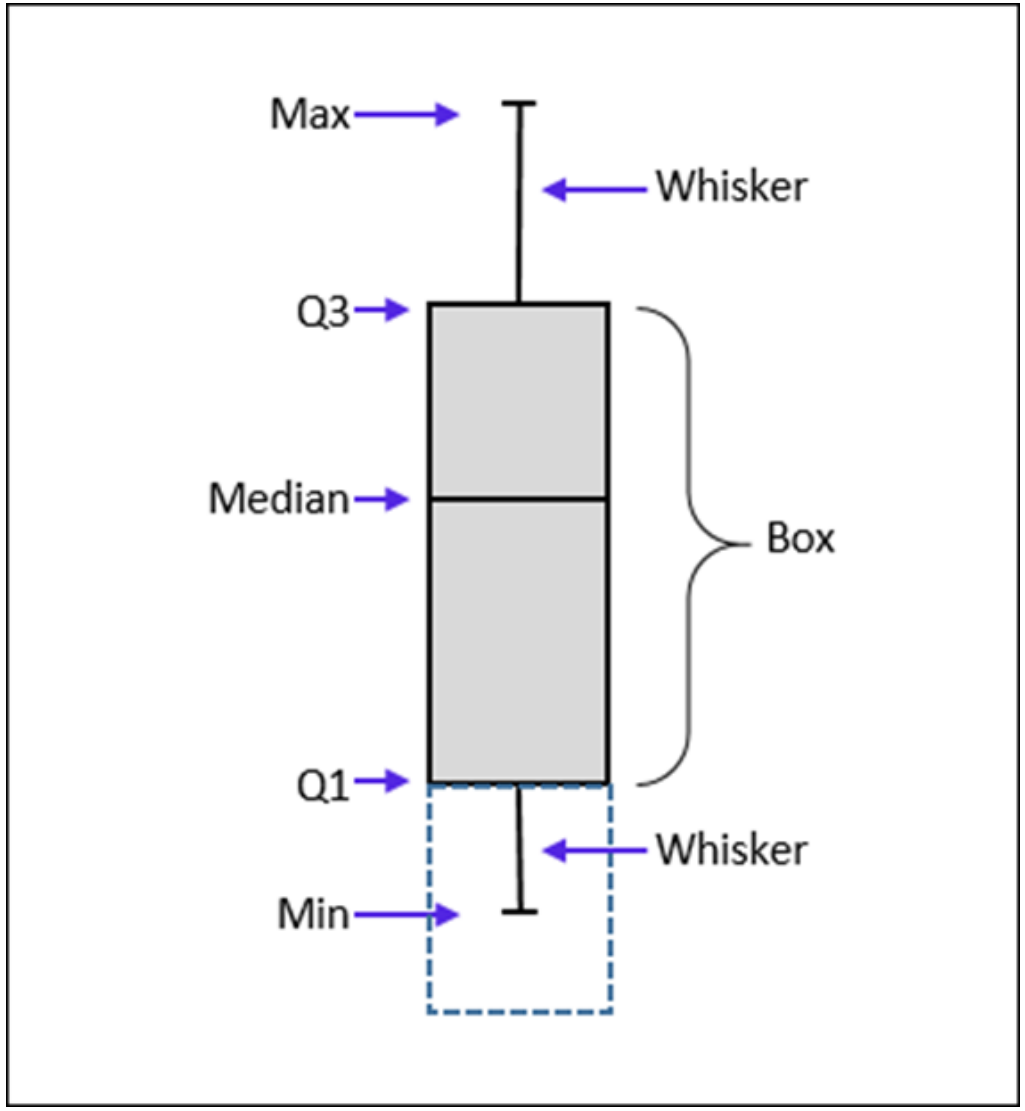
Measuring The Dispersion of Data

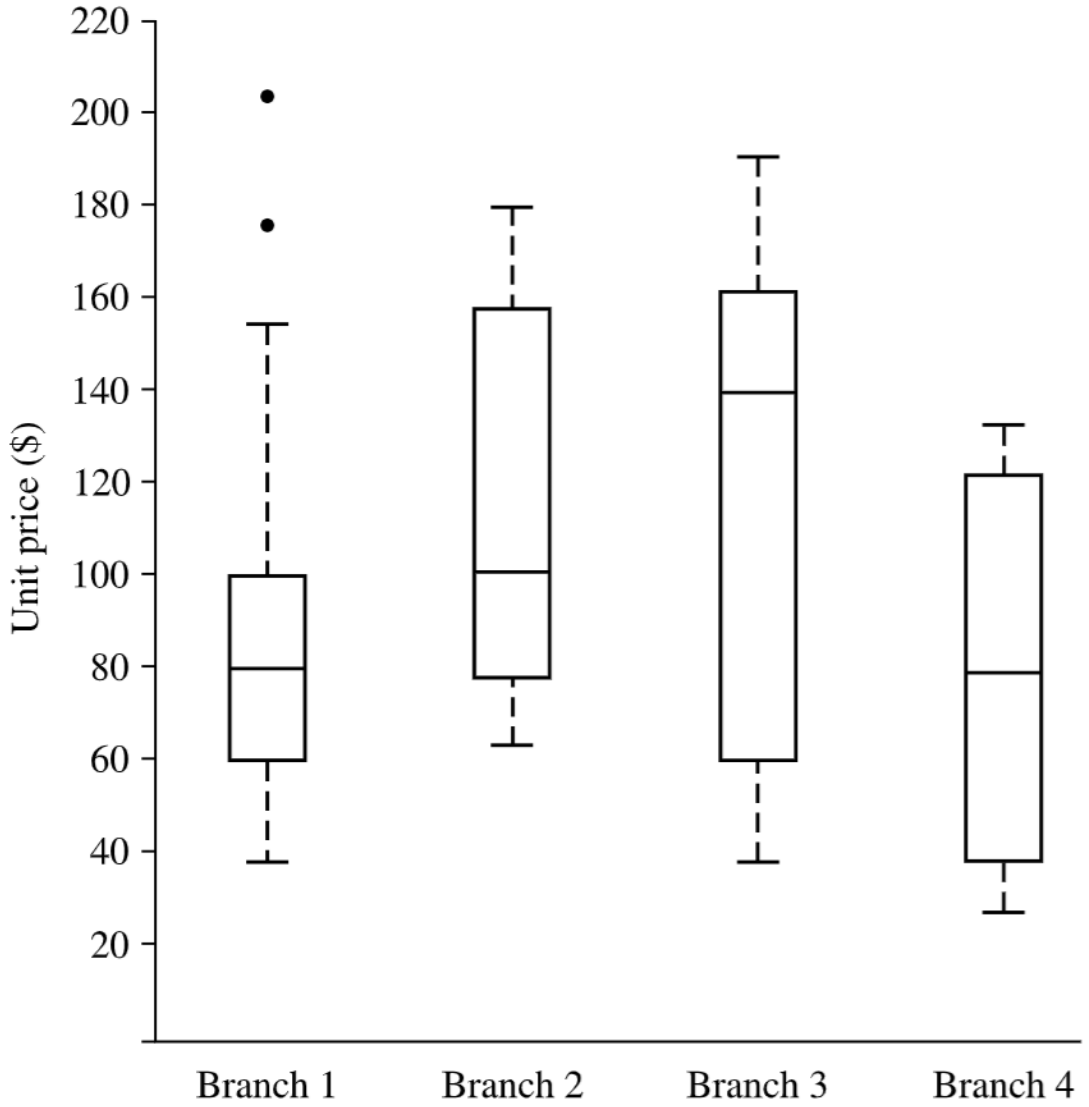
- Percentiles (100-quantiles)
  - Percentiles divide ordered data into 100 approximately equal-sized groups using 99 cut points.
- Interquartile range
  - $IQR = Q_3 - Q_1$
  - Potential outliers: 1.5 × IQR above the third quartile or below the first quartile
    - Lower: $Q_1 - 1.5 \times IQR$
    - Upper: $Q_3 + 1.5 \times IQR$

Five-Number Summary and Boxplot

- Five-number summary (五數概括法)
  - The five-number summary describes the center, spread, and range of a dataset.
  - Minimum, Q_1 , Median, Q_3 , Maximum
- Boxplot: Data is represented with a box
  - The box extends from Q_1 to Q_3.
  - The line inside the box represents the median.
  - The box length represents the IQR.
  - Whiskers extend to the most extreme non-outlier values.
  - Potential outliers are plotted individually.

Boxplot


Max
Whisker
Q3
Median
Box
Q1
Whisker
Min

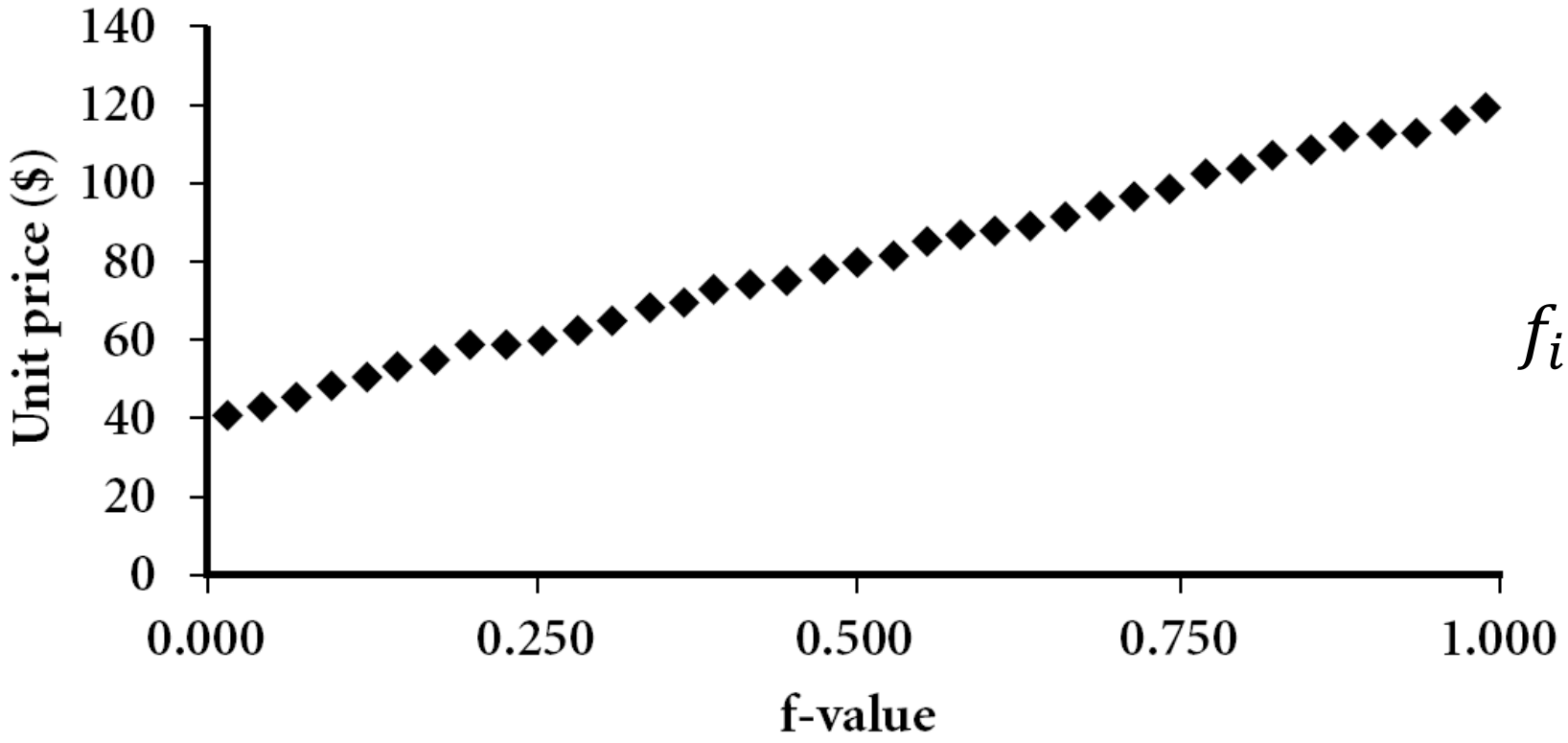

220
200
180
160
140
120
100
80
60
40
20
Unit price ($)
Branch 1
Branch 2
Branch 3
Branch 4

③ Graphical Displays of Data

- Boxplots
  - Summarize center, spread, and potential outliers
- Quantile plots
  - Show ordered values against cumulative proportions
- Q–Q plots
  - Compare two distributions through their corresponding quantiles
- Histograms
  - Show the distribution of one numerical variable
- Scatter plots
  - Show the relationship between two numerical variables

Quantile Plot

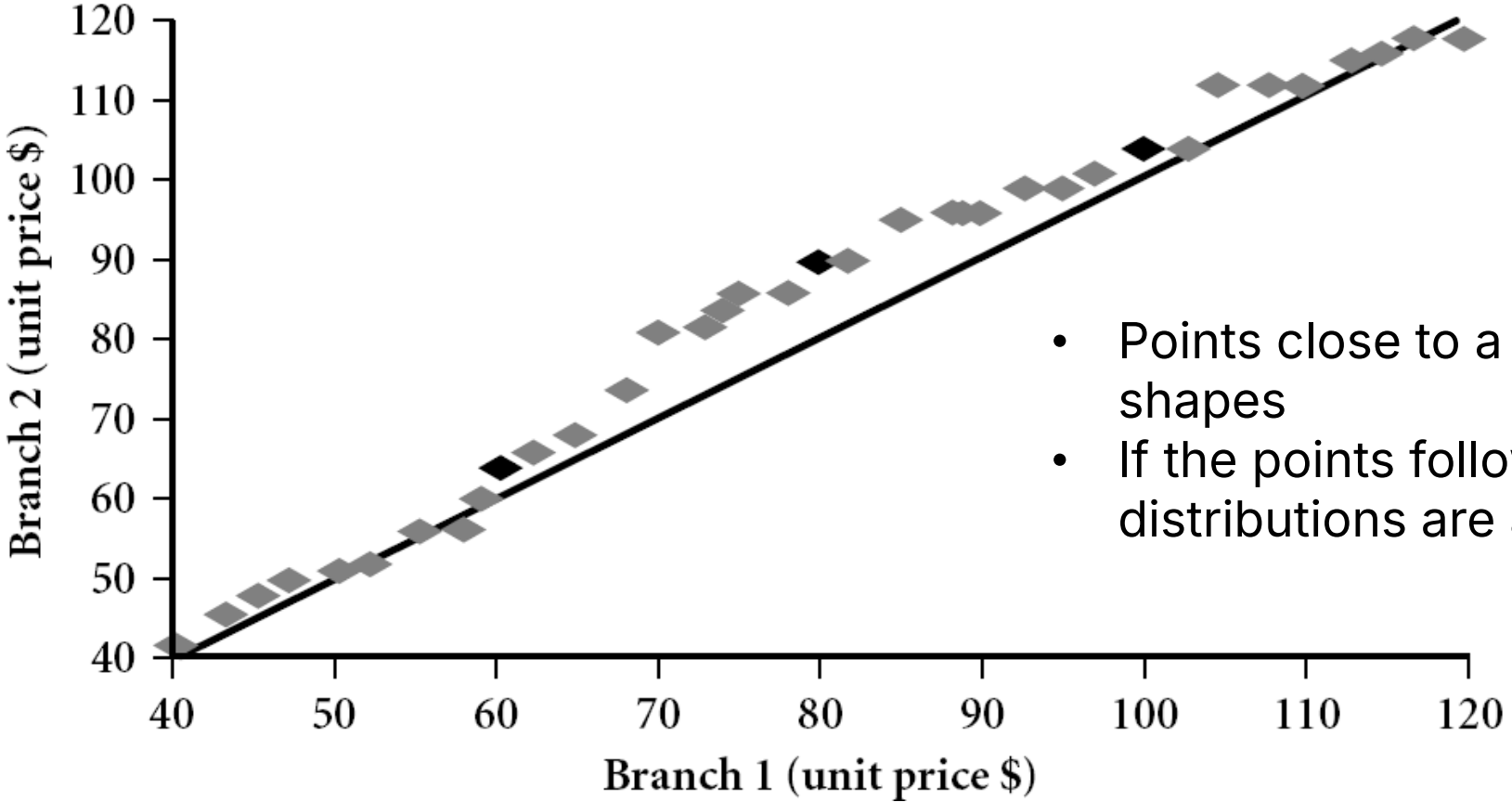
- A quantile plot displays each ordered observation $x_{(i)}$ against its plotting position f_i



$$f_i = \frac{i - 0.5}{N}, i = 1, \ldots, N$$

Quantile-Quantile (Q-Q) Plot

- A Q–Q plot compares corresponding quantiles from two distributions.

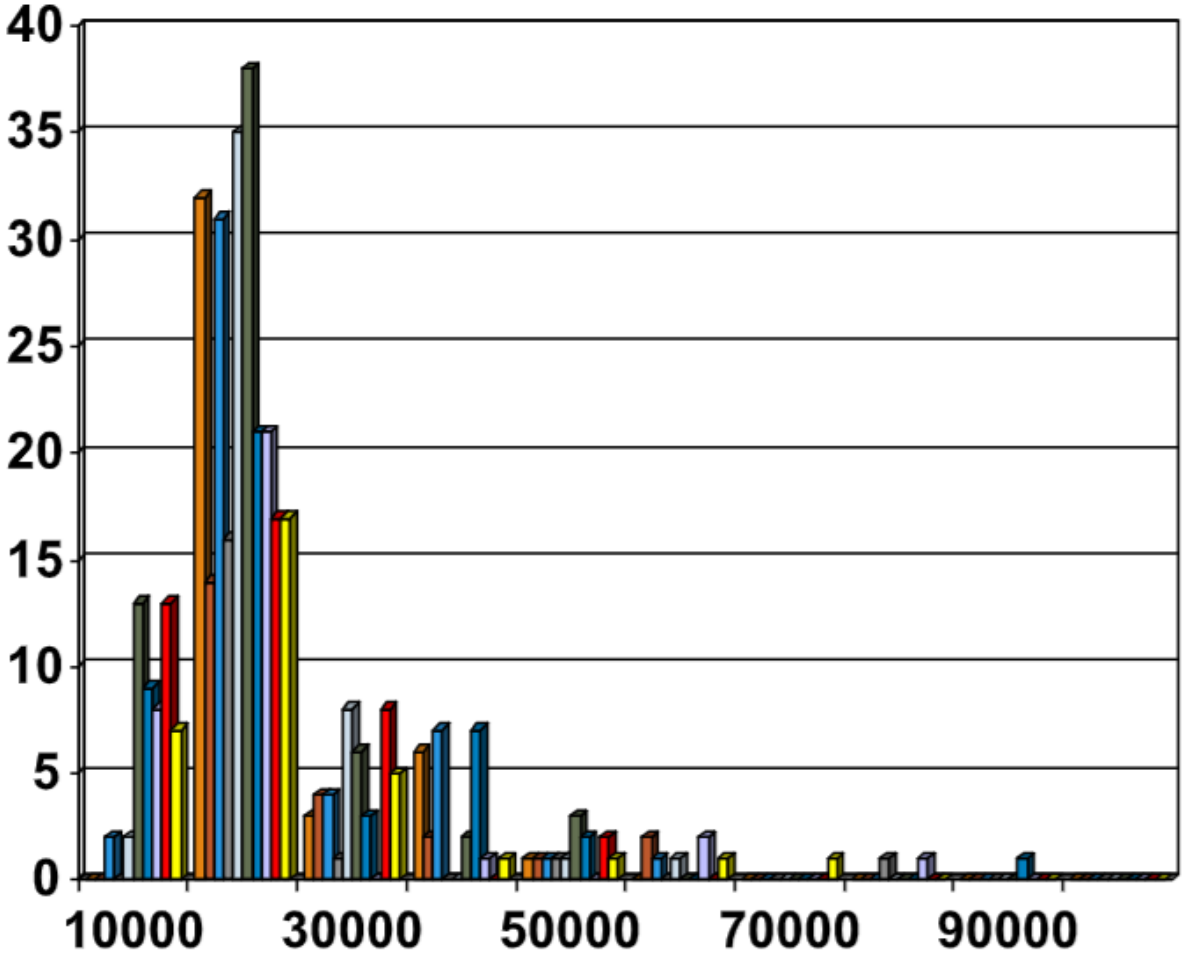


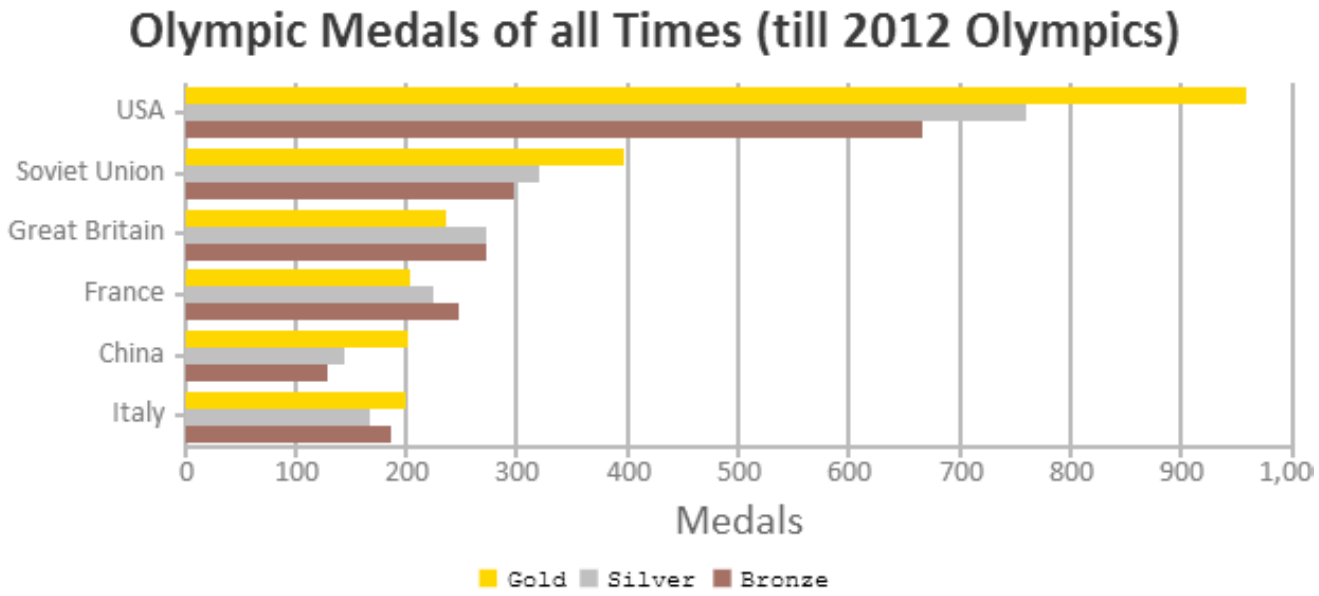
- Points close to a straight line: similar distributional shapes
- If the points follow the reference line y=x, the two distributions are approximately the same.

Histograms

- A histogram divides numerical data into consecutive intervals called bins and shows the frequency or density within each bin.
- Histogram versus Bar chart
  - Histograms show the distribution of a numerical variable, whereas bar charts compare values across categories.

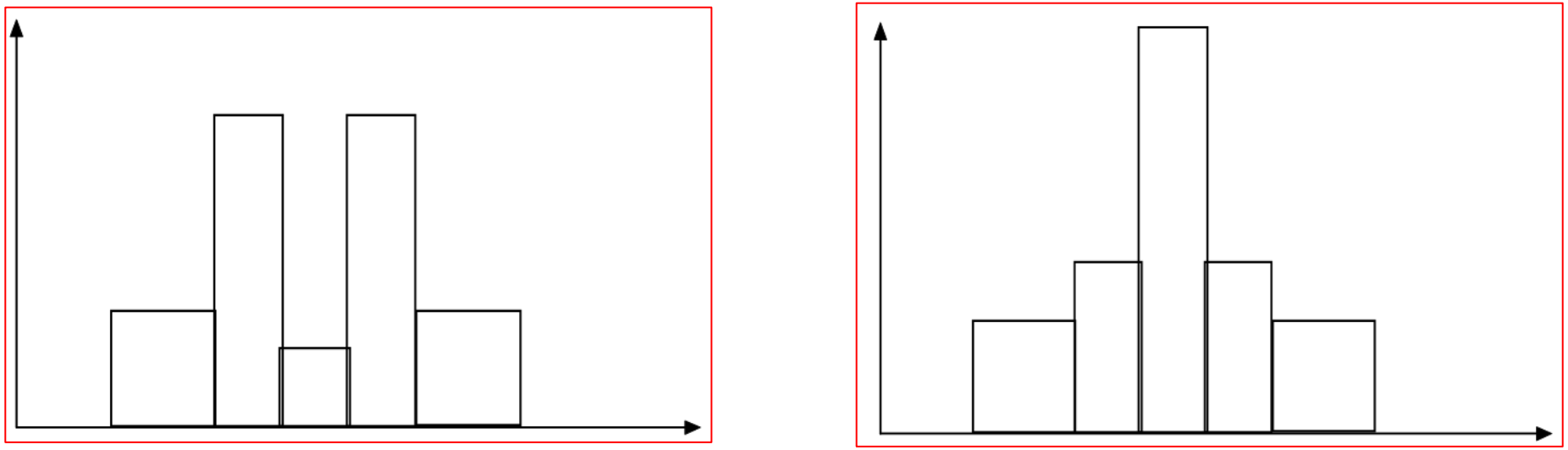
Histogram	Bar chart
Numerical data	Categorical data
Bins represent intervals	Bars represent categories
Bin order has numerical meaning	Categories may often be reordered
Adjacent bars usually touch	Bars are usually separated
Shows distribution shape	Compares category values


40
35
30
25
20
15
10
5
0
10000
30000
50000
70000
90000


Olympic Medals of all Times (till 2012 Olympics)
USA
Soviet Union
Great Britain
France
China
Italy
0
100
200
300
400
500
600
700
800
900
1,00
Medals
Gold
Silver
Bronze

Histogram vs. Boxplot

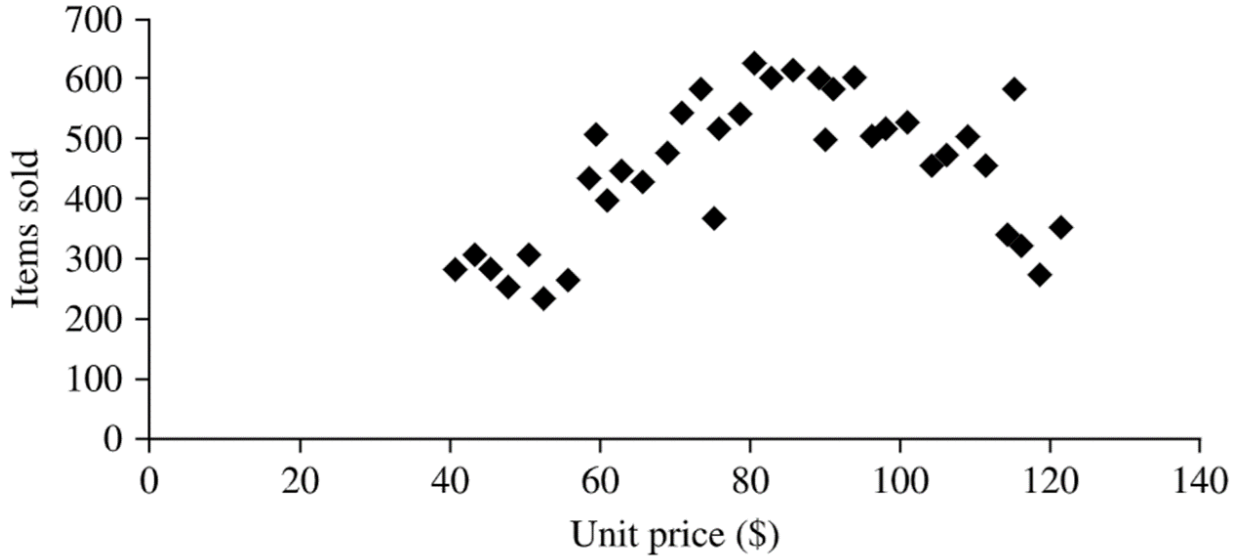
- Histograms sometimes tell more than Boxplots
  - Different distributions may share the same five-number summary.
  - Therefore, they may produce identical or very similar boxplots.



Two very different distributions can have the same five-number summary and boxplot.

Scatter plots

- A scatter plot represents each paired observation (x_i, y_i) as a point in a two-dimensional plane.
- It helps reveal the direction, strength, shape, clusters, and outliers in a bivariate relationship.
- Provides a first look at bivariate data



Data quality, cleaning, integration, transformation and reduction

Major Tasks in Data Preprocessing

① Data cleaning
- Handle missing values and noisy data.
- Detect outliers and correct inconsistent or duplicate records.

② Data integration
- Combine data from multiple sources
- Resolve schema, naming, unit, and entity conflicts

Major Tasks in Data Preprocessing

③ Data transformation & Data discretization
- Scaling, normalization, and feature construction
- Convert continuous values into discrete intervals

④ Data reduction
- Reduce data volume while preserving essential information
- Sampling, feature selection, dimensionality reduction



- Data Cleaning
Fill in missing values, smooth noisy data, identify or remove outliers, and correct inconsistencies



- Data Transformation
Normalization and aggregation

- Data Integration
Integration of multiple databases, data cubes, or files



- Data Reduction
Part of data reduction but with particular importance, especially for numerical data

Source: Impact of data

Data Quality Issues

1. Accuracy: Are the values correct? 芙莉蓮年紀 age = 18
2. Completeness: Are required values available? 空白的100年
3. Consistency: Do values agree across records or sources? 江戶川柯南：7歲 /工藤新一：17歲
4. Timeliness: Are the data sufficiently up to date? Move house
5. Believability: Are the data considered credible and trustworthy? 紅色三倍速
6. Interpretability: Are the meaning, units, and format clear? My cat’s weight = 6000

① Data Cleaning

- Real-world data are often dirty because of faulty instruments, human errors, software errors, or transmission failures.
  - Incomplete:
    - Required values, attributes, or records are missing.
    - Occupation = NULL
  - Noisy: containing noise, errors, or outliers.
    - Values contain random errors or unwanted variations.
    - A temperature sensor reports 25.1, 25.0, 79.8, 25.2 °C
  - Inconsistent: containing discrepancies in codes or names.
    - Age = “42”, Birthday = “03/07/2010”
    - Was rating “1, 2, 3”, now rating “A, B, C”
    - discrepancy between duplicate records
  - Intentional: disguised missing data.
    - Feb. 29 as everyone’s birthday?

Missing Data

- Some attribute values or entire records may be unavailable.
- Common causes include:
  a) Equipment or sensor failure
  b) Human entry errors or skipped fields
  c) Values removed after validation
  d) Attributes not collected because they were considered unimportant
  e) Failures in data transmission or system logging
- The appropriate treatment depends on why the data are missing.

How to Handle Missing Data?

1) Delete the record
   - Only a small proportion of records are missing.
   - Missingness is approximately random.
   - Removing the records does not introduce serious bias.
2) Fill in the missing value manually
   - May provide accurate values when domain experts are available.
   - Time-consuming, subjective, and impractical for large datasets.

How to Handle Missing Data? (cont.)

3) Fill in it automatically with

- Global constant to fill in the missing value: unknown
- Simple statistical imputation
  - Numerical attributes: mean or median
  - Categorical attributes: mode
- Group-wise imputation
  - Use the mean, median, or mode within a meaningful group.
- Model-based imputation
  - Regression
  - Decision trees
  - k-nearest neighbors

Fit the imputation method using the training set only.

Noisy Data

- Noisy data contain random errors or unwanted variations that distort the underlying true value or pattern.
  - Observed value = True value + Noise
- Common causes:
  - Environmental interference
  - Data entry errors
  - Data transmission errors
  - Limited measurement precision or quantization

Unusual variation is not necessarily noise.

How to Handle Noisy Data?

- Binning
  - First sort data and partition into (equal-width / equal-frequency) bins
  - Then one can smooth by bin means, smooth by bin median, smooth by bin boundaries, etc.
- Clustering
  - Detect potential outliers as objects that do not belong strongly to any cluster.
- Regression
  - Fit a regression function to capture the underlying trend and identify large residuals.

Equal-width vs. Equal-frequency

- Equal-width binning
  - Divides the range into N intervals of equal size.
  - The width of intervals: $\frac{\max(X)-\min(X)}{\mathrm{N}}$
  - Outliers may greatly expand the range and produce poorly populated bins.
  - Skewed data is not handled well.
- Equal-frequency partitioning
  - Sorts the observations and divides them into N bins containing approximately the same number of observations.
  - Bin widths may be very different.
  - Nearby values may be separated by a bin boundary.

Binning Example

- Sorted data for price
  - 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34
- Partition into equal-frequency bins (N=3):
  1) Bin 1: 4, 8, 9, 15
  2) Bin 2: 21, 21, 24, 25
  3) Bin 3: 26, 28, 29, 34

Smoothing by bin means:
- Bin 1: 9, 9, 9, 9
- Bin 2: 23, 23, 23, 23
- Bin 3: 29, 29, 29, 29

Smoothing by bin boundaries:
- Bin 1: 4, 4, 4, 15
- Bin 2: 21, 21, 25, 25
- Bin 3: 26, 26, 26, 34

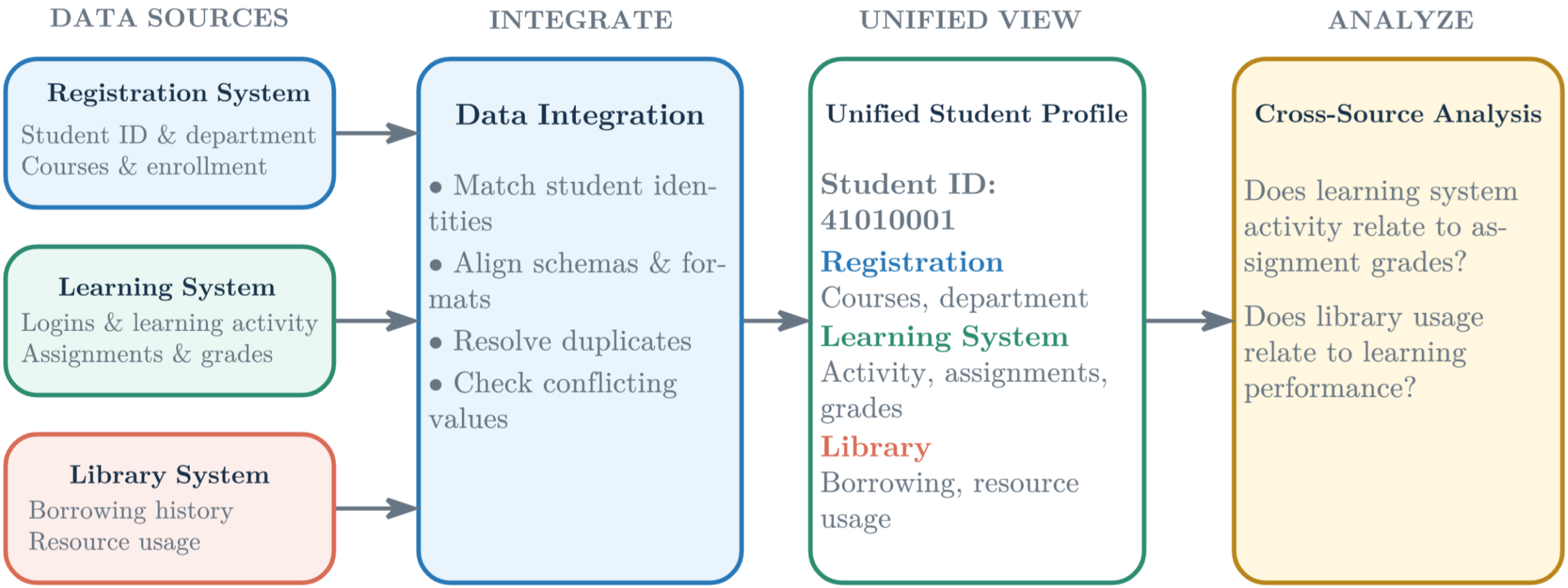
② Data Integration

- Combines data from multiple sources into a unified and consistent view
- Why data integration?
  - Provide a more complete view of the entities being analyzed
  - Enable analysis across systems and data sources
  - Improve analytical coverage and support better decisions

② Data Integration (cont.)

- Common challenges
  - Schema and naming conflicts:
    - student_id vs. sid
  - Different units and formats:
    - 70 kg vs. 154 lb
  - Duplicate records:
    - The same student appears twice.
  - Entity identification problems:
    - 41010001 vs. Jason Huang
  - Conflicting attribute values:
    - Different departments for the same student.

Example



Separate systems provide partial views; integration enables a more complete analysis.

Handling Conflicting Values

- Detecting value conflicts
  - Different sources may report different values for the same entity and attribute.
  - Possible reasons: no reason, different representations, different scales, e.g., metric vs. British units
- Resolve value conflicts
  - Prefer an authoritative or reliable source
  - Prefer the most recent value when appropriate
  - Preserve multiple values if the conflict cannot be resolved

Do not force a single value when there is insufficient evidence.

Handling Redundancy

- Redundancy may arise when multiple sources describe the same entities or information.
  - Entity identification
    - The same attribute may have different names.
    - The same entity may have different identifiers.
  - Derivable data
    - One attribute may be a “derived” attribute in another table
    - $\text{Annual revenue} = \sum_{m=1}^{12} \text{Monthly revenue}_{\text{m}}$
  - Redundant attributes may be detected by correlation analysis and covariance analysis

Highly correlated attributes are not necessarily redundant.

③ Data Transformation

- Data transformation converts data into forms that are more suitable for analysis and modeling.
- Common methods
  - Feature construction: Create new features from existing attributes
  - Normalization: Rescale numerical values to a specified range
  - Standardization: Center and scale numerical values
  - Discretization: Convert continuous values into intervals or categories
  - Concept hierarchy generation: Replace low-level values with higher-level concepts

Normalization and Standardization

- Min-max normalization: x_i in the range
  - $[a, b]$: $x_i' = \frac{x_i - \min(X)}{\max(X) - \min(X)}(b - a) + a$
  - $[0,1]$: $x_i' = \frac{x_i - \min(X)}{\max(X) - \min(X)}$
- Z-score standardization
  - $z_i = \frac{x_i - \mu_X}{\sigma_X} \Rightarrow mean(Z) = 0, std(Z) = 1$
- Normalization by decimal scaling
  - $x'_i = \frac{x_i}{10^j}$, where j is the smallest integer such that $\max(|x'_i|) < 1$

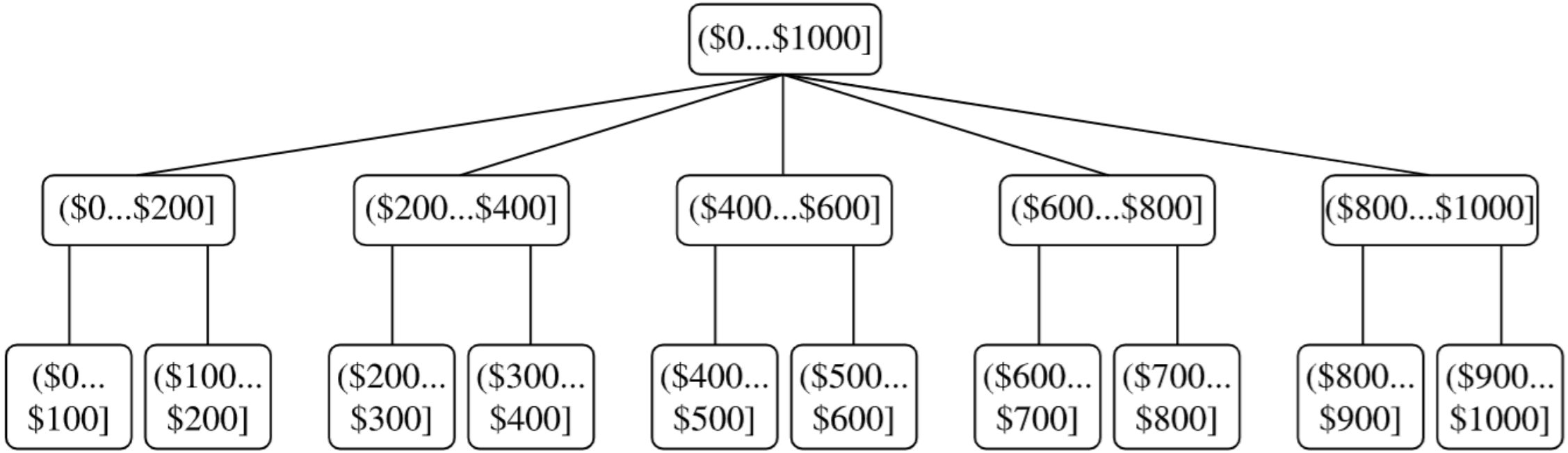
Estimate scaling parameters from the training set only

Discretization

- Discretization converts a continuous numerical attribute into a finite set of intervals or categories.
  - Rhythm Game: Accuracy → {C,B,A,S,SS}
- Simple discretization: Binning
  - Equal-width binning: All intervals have the same width
  - Equal-frequency binning: All bins contain approximately the same number of observations
- Possible uses
  - Produce categorical features
  - Simplify patterns and rules
  - Reduce sensitivity to small numerical variations

Concept hierarchies for Numerical Data

- A numerical concept hierarchy replaces detailed numerical values with increasingly general intervals or categories.



Concept hierarchy for Categorical data

- A concept hierarchy organizes detailed categorical values into progressively more general concepts.
- Attribute hierarchy
  - Street ➔ City ➔ State ➔ Country
- Value grouping
  - {Los Angeles Lakers , San Antonio Spurs, Golden State Warriors} ➔ WEST
  - {Boston Celtics, Miami Heat, Chicago Bulls} ➔ EAST
  - {WEST, EAST} ➔ NBA

④ Data Reduction

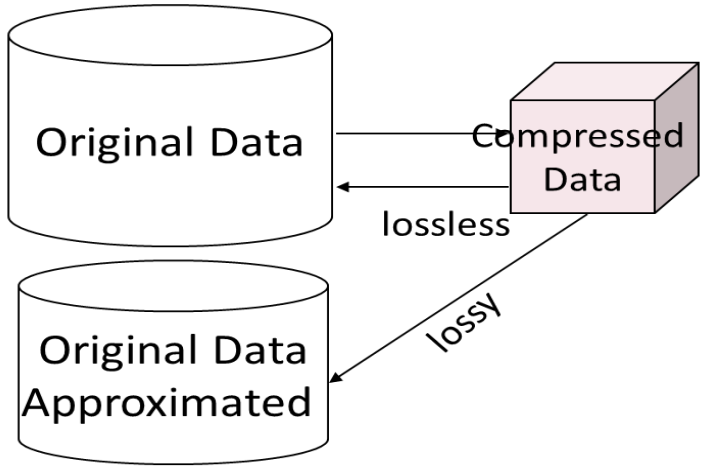
- Produce a smaller representation of a dataset while preserving most of the information needed for analysis.
- Why reduce data?
  - Reduce storage and memory requirements
  - Speed up data mining and model training
  - Simplify analysis and visualization
  - Retain approximately the same analytical conclusions

④ Data Reduction (cont.)

- Common strategies
  - Data compression: Encode data using fewer bits
    - Lossless or lossy compression
  - Sampling: Reduce the number of data objects
  - Dimensionality reduction: Reduce the number of attributes
    - PCA, Feature Selection

Data Compression

- Represent data using fewer bits to reduce storage and transmission costs.
  - Lossless compression
    - Reconstructs the original data exactly
    - Examples: ZIP, gzip, PNG, FLAC
    - Common for text and structured data
  - Lossy compression
    - Discards some information for a higher compression ratio
    - Examples: JPEG, MP3, AAC, H.264
    - Common for perceptual multimedia data



Lossy vs. lossless compression

Sampling

- Sampling selects a smaller subset S from a population or dataset D to estimate properties of the full dataset.

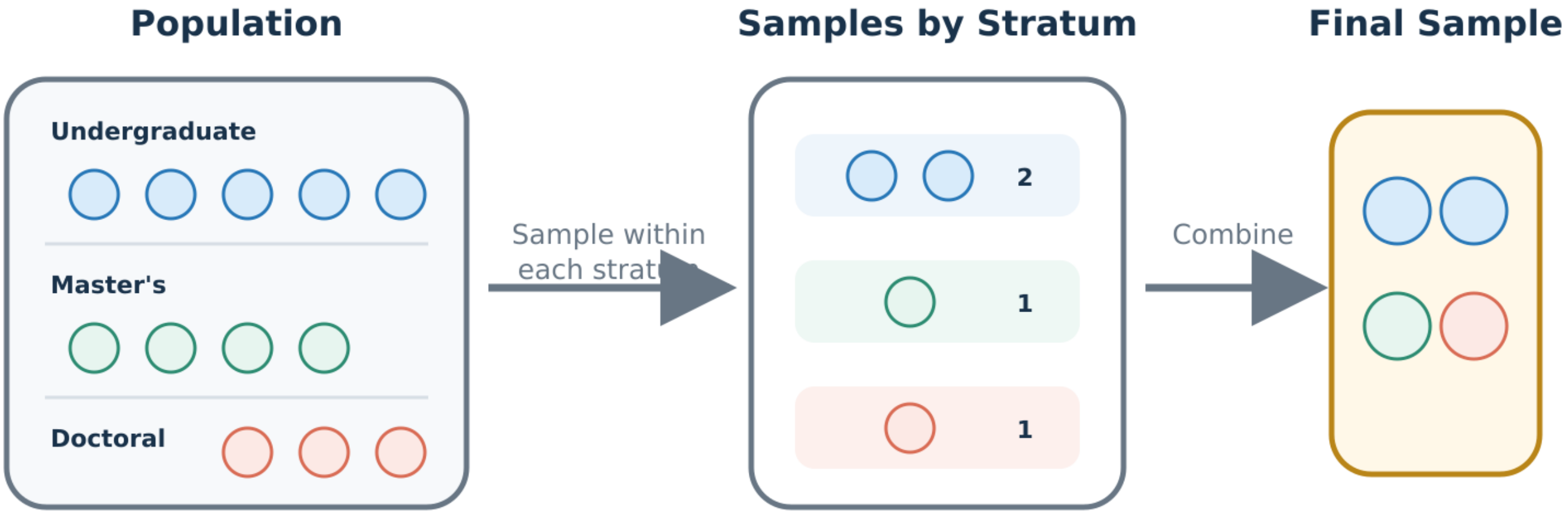
$$S \subseteq D, |S| = s \ll N = |D|$$

- Running an algorithm on the sample can substantially reduce computation time and memory usage.
  - Choose a representative subset of the data
- Potential problem
  - Small random samples may miss rare or minority groups
  - Biased sampling produces biased conclusions

Sampling (cont.)

- Simple random sampling: Every object has an equal probability of selection.
  - Sampling without replacement
    - A selected object can be selected again.
  - Sampling with replacement
    - A selected object cannot be selected again.
- Stratified sampling
  - Divide the population into meaningful, non-overlapping strata and sample from every stratum.
    - Students → {Undergraduate, Master, PhD}

Stratified Sampling Example



Every subgroup is represented in the final sample

What is Dimensionality Reduction?

- When dimensionality increases, data becomes increasingly sparse
  - Represents data using fewer features while preserving task-relevant information

$$\mathbb{R}^d \rightarrow \mathbb{R}^k, k < d$$

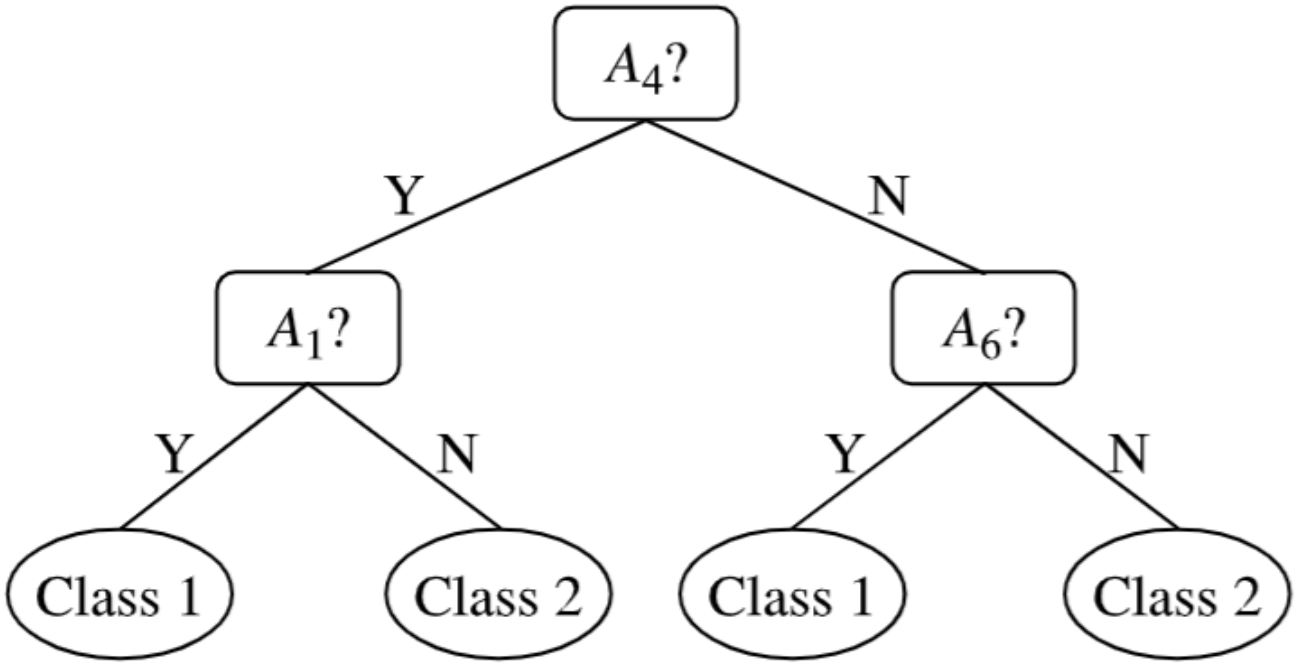
- Why reduce dimensionality?
  - Remove potentially irrelevant or redundant information
  - Reduce computation and storage
  - Enable 2-D or 3-D visualization

Dimensionality Reduction Methods

- Dimensionality reduction methodologies
  - Feature selection: Select a subset of the original features.
    - $\{x_1, x_2, \dots, x_d\} \rightarrow \{x_2, x_5, x_8\}$
  - Feature extraction: Transform the original features into a lower-dimensional representation. (New transformed features)
    - $z = W^T x, z \in \mathbb{R}^k, x \in \mathbb{R}^d$
- Some typical dimensionality reduction methods
  - Attribute Subset Selection
  - Principal Component Analysis (PCA)

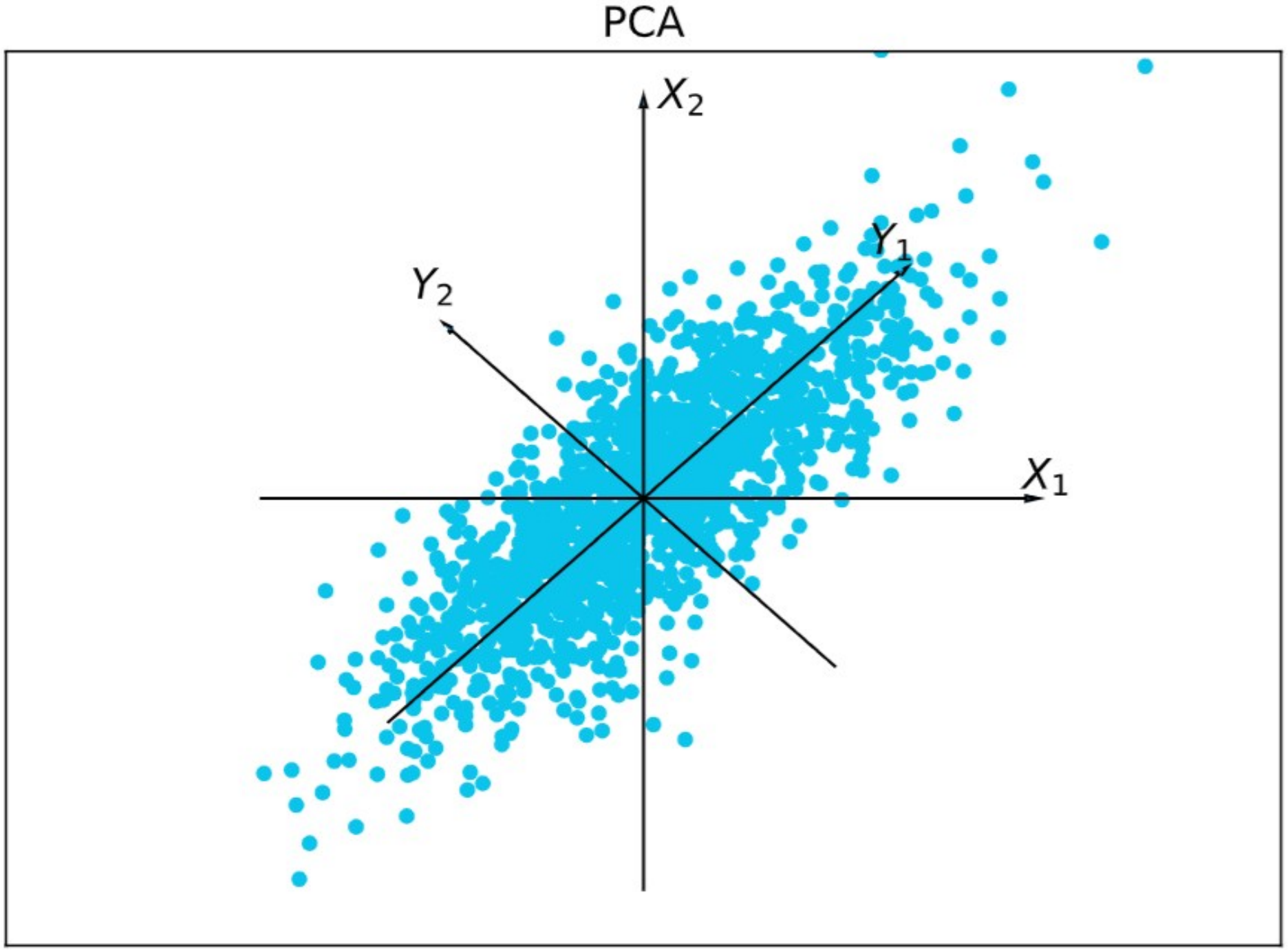
Attribute Subset Selection

- Typical heuristic attribute selection methods:
  - Step-wise feature selection:
    - The best single-attribute is picked first
    - Then next best attribute condition to the first, ...
  - Step-wise attribute elimination:
    - Repeatedly eliminate the worst attribute
  - Combined selection and elimination
  - Decision-tree induction

Forward selection	Backward elimination	Decision tree induction
Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$ Initial reduced set: $\{\}$ => $\{A_1\}$ => $\{A_1, A_4\}$ => Reduced attribute set: $\{A_1, A_4, A_6\}$	Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$ => $\{A_1, A_3, A_4, A_5, A_6\}$ => $\{A_1, A_4, A_5, A_6\}$ => Reduced attribute set: $\{A_1, A_4, A_6\}$	Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$  => Reduced attribute set: $\{A_1, A_4, A_6\}$

Principal Component Analysis

- Given n observations with d numerical features, PCA finds $k \leq d$ orthogonal directions that capture as much variance as possible.
  - The first axis shows the most variance among the data, the second axis shows the next highest variance, and so on.
- PCA preserves variance, not necessarily information useful for predicting a target.


PCA
X_2
Y_1
Y_2
X_1

Principal Component Analysis (cont.)

- Procedure:
  - Center the data
  - Compute the covariance matrix
  - Compute eigenvectors and eigenvalues
    - Eigenvectors define the principal directions.
    - Eigenvalues indicate the variance captured by those directions.
- Standardize features before PCA when their units or scales differ substantially.
- Standard PCA operates on a numerical data matrix.